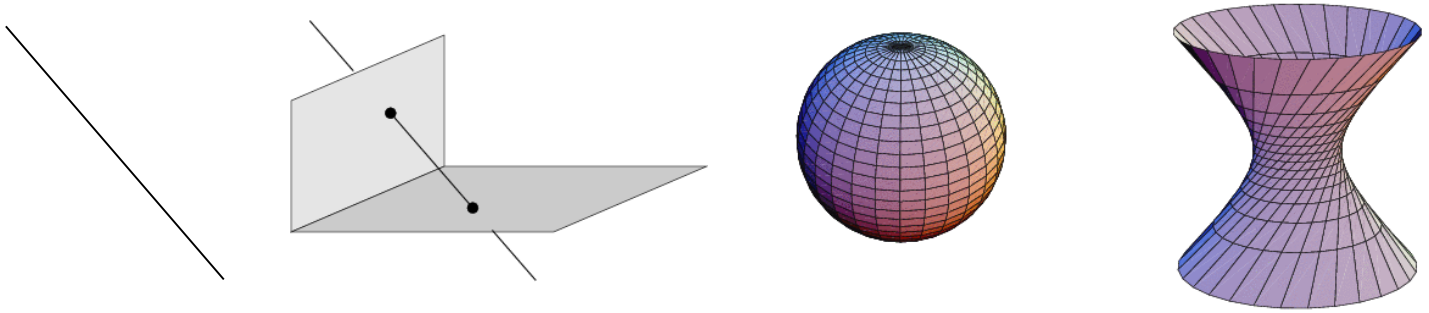
- Introduction
- Planar (2D) Projective Geometry
- **Space (3D) Projective Geometry**
- Camera Geometry (3D → 2D Projection)

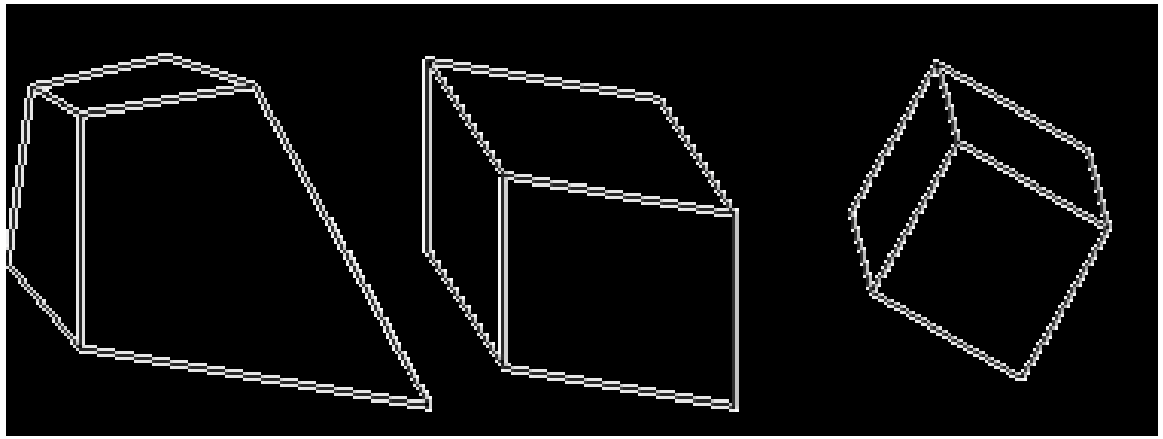
Space (3D) Projective Geometry

3D Space Projective Geometry

- **Elements**
- **Points**
- Planes
- Quadrics
- (Dual quadrics)

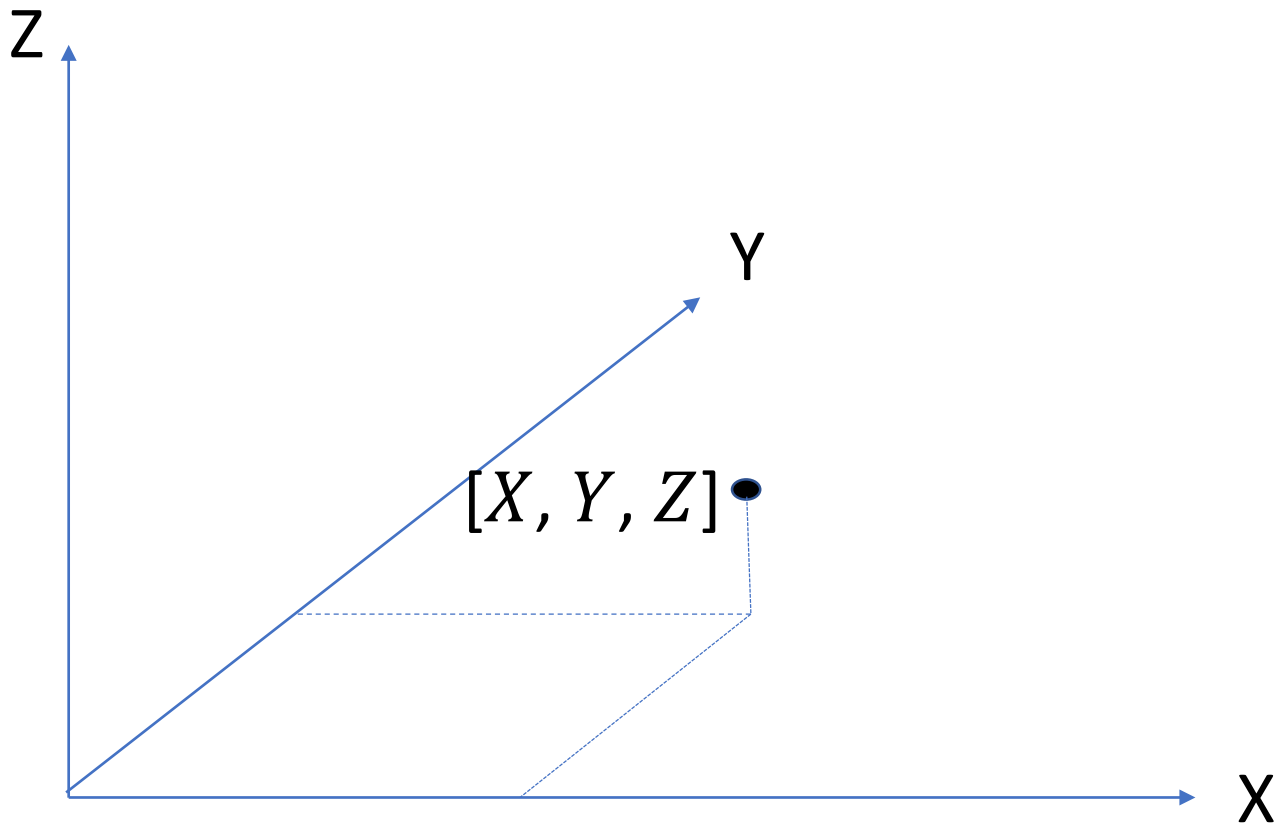


- **Transformations**
- Isometries
- Similarities
- Affinities
- Projectivities

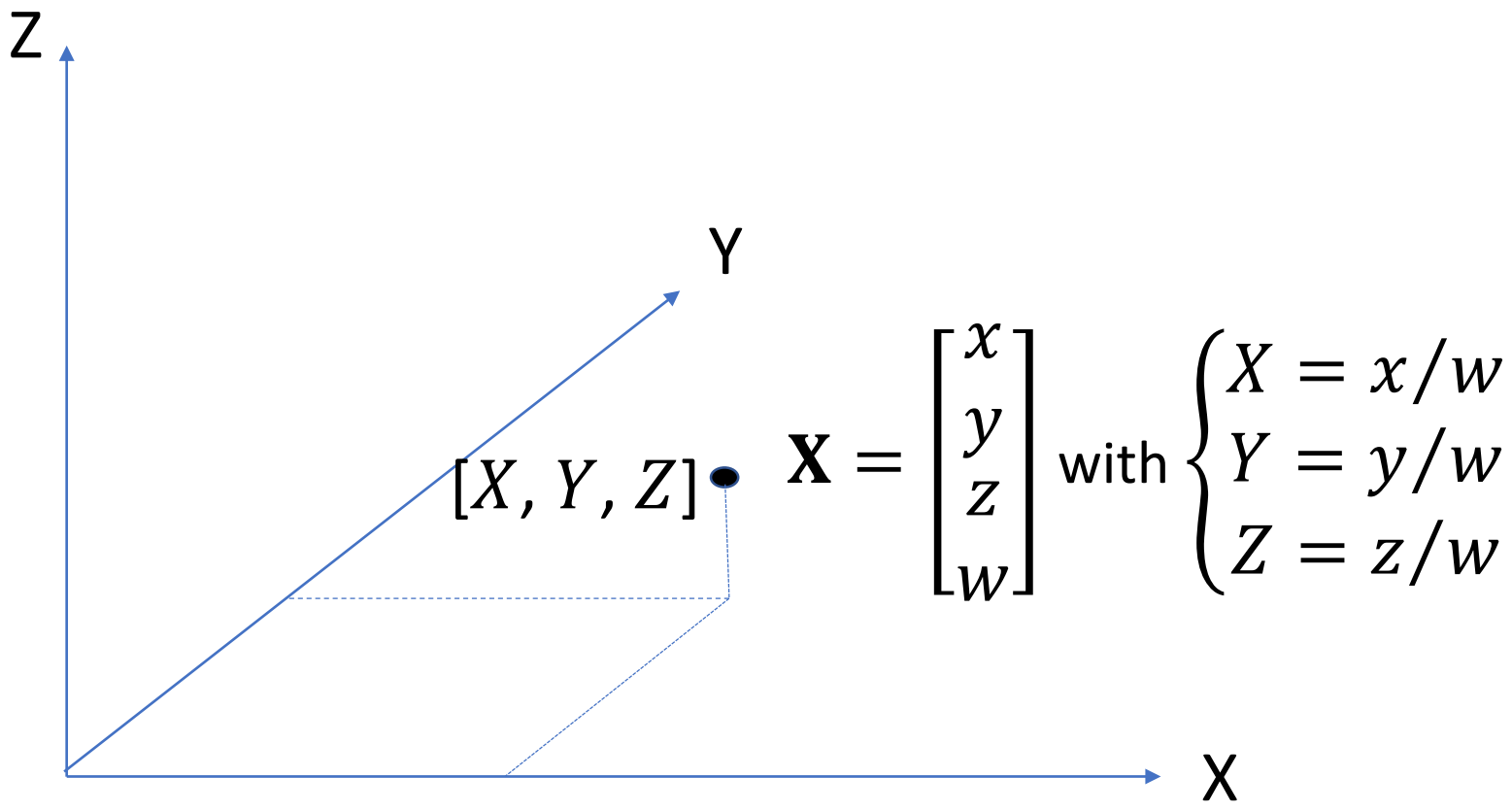


Points in the projective space

Euclidean space (3D) cartesian coordinates

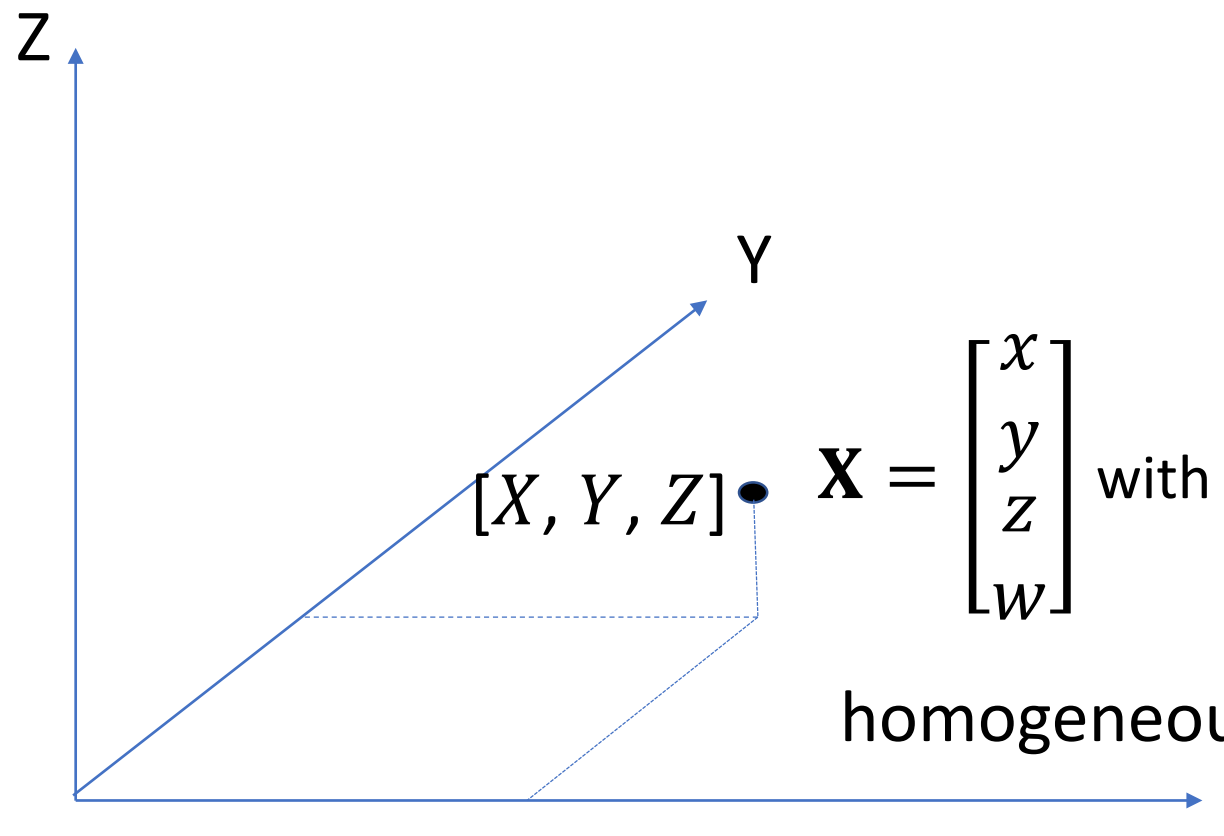


Projective space (3D)
4 homogeneous coordinates



$$\mathbf{X} = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \text{ with } \begin{cases} X = x/w \\ Y = y/w \\ Z = z/w \end{cases}$$

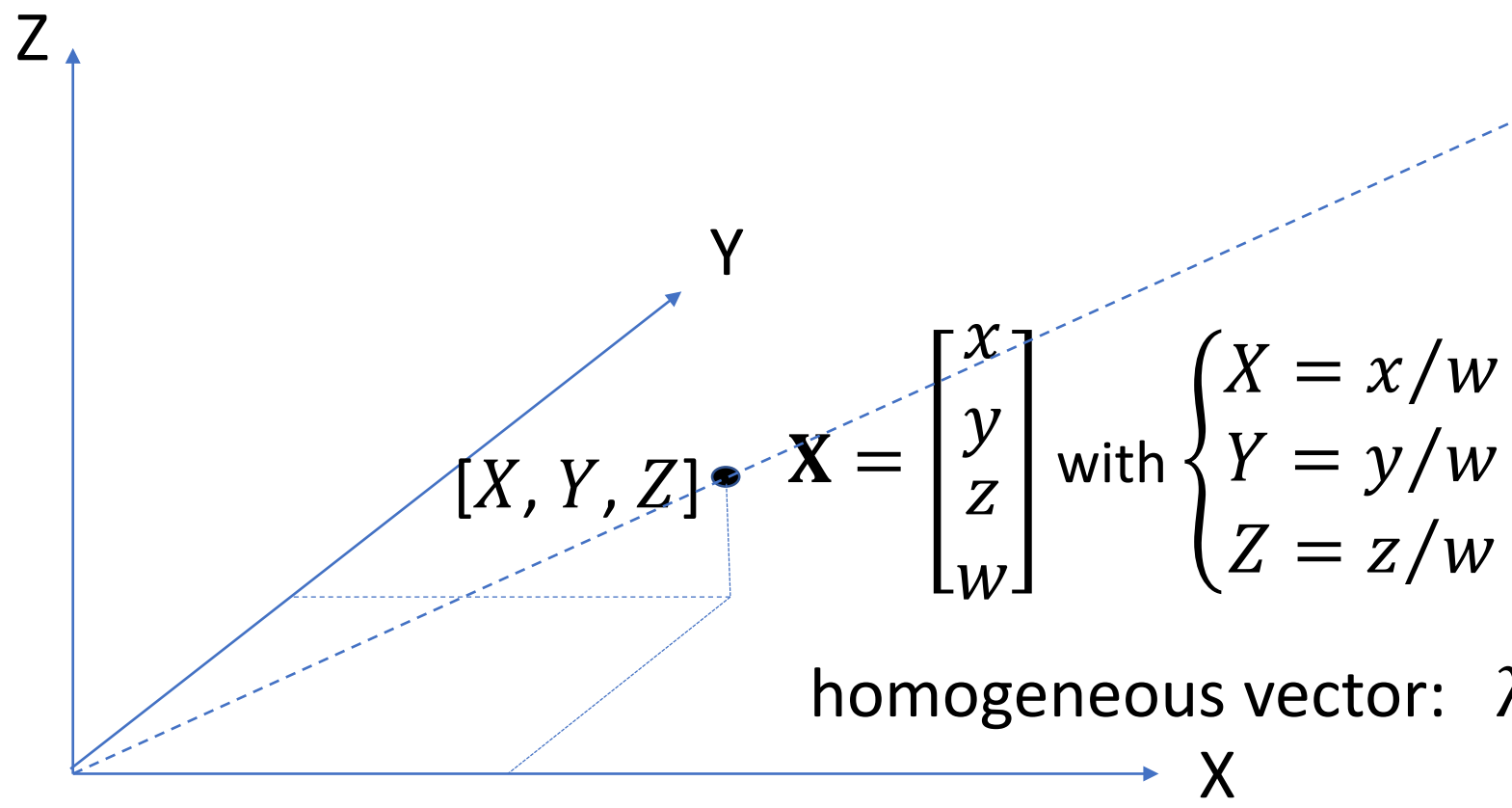
Projective space (3D)
4 homogeneous coordinates



$$\mathbf{X} = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \text{ with } \begin{cases} X = x/w \\ Y = y/w \\ Z = z/w \end{cases}$$

homogeneous vector: $\lambda\mathbf{X} \Leftrightarrow \mathbf{X}$

Projective space $\mathbb{P}^3$: points at the ∞



$$\mathbf{X}_\infty = \begin{bmatrix} x \\ y \\ z \\ 0 \end{bmatrix}$$

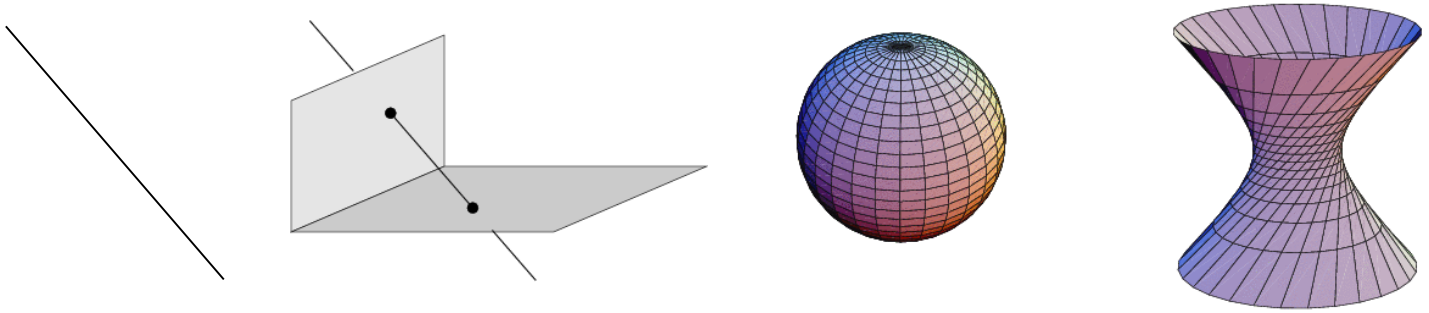
$$\mathbf{X} = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \text{ with } \begin{cases} X = x/w \\ Y = y/w \\ Z = z/w \end{cases}$$

homogeneous vector: $\lambda\mathbf{X} \Leftrightarrow \mathbf{X}$

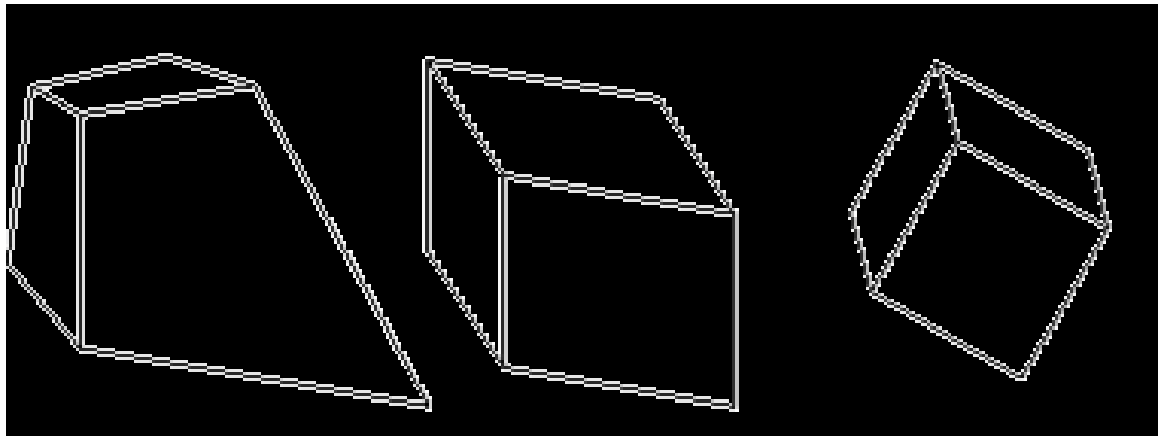
$$\mathbb{P}^3 = \{\mathbf{X} \in \mathbb{R}^4 - \{[0 \quad 0 \quad 0 \quad 0]^T\}\}$$

3D Space Projective Geometry

- **Elements**
- Points
- **Planes**
- Quadrics
- (Dual quadrics)

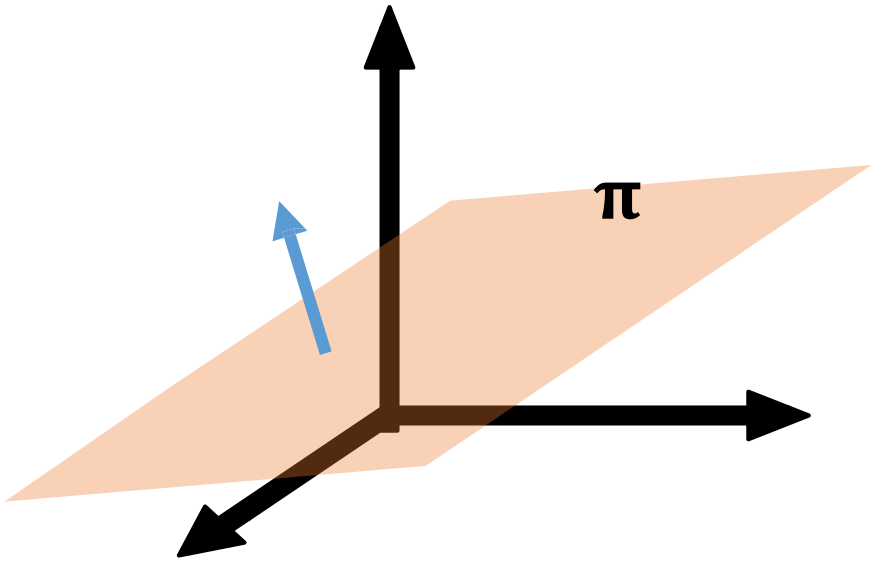


- **Transformations**
- Isometries
- Similarities
- Affinities
- Projectivities



Planes in the projective space

Planes in 3D Projective Geometry

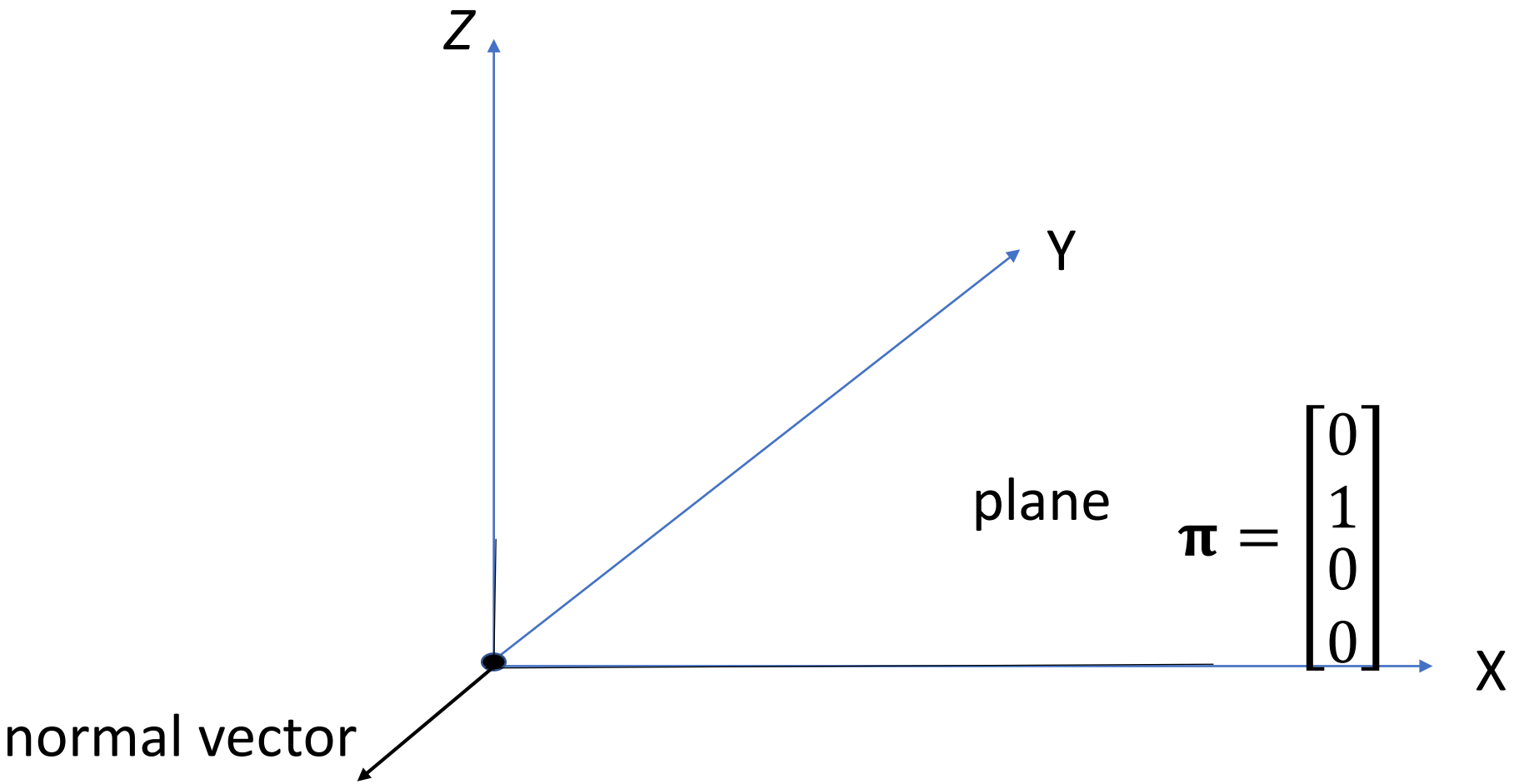


$\boldsymbol{\pi} = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$ with direction $\begin{pmatrix} a & b & c \end{pmatrix}$ normal to the plane,

and $\dfrac{-d}{\sqrt{a^2+b^2+c^2}}$ is the distance between the origin and the plane

$\boldsymbol{\pi}$ is a homogeneous vector: $\lambda\boldsymbol{\pi} \Leftrightarrow \boldsymbol{\pi}$

Example: the X-Z plane



The incidence relation:
a point is on a plane, or a plane goes through a point

$$ax + by + cz + dw = 0$$
$$\boldsymbol{\pi}^T\mathbf{X} = 0 = \mathbf{X}^T\boldsymbol{\pi}$$

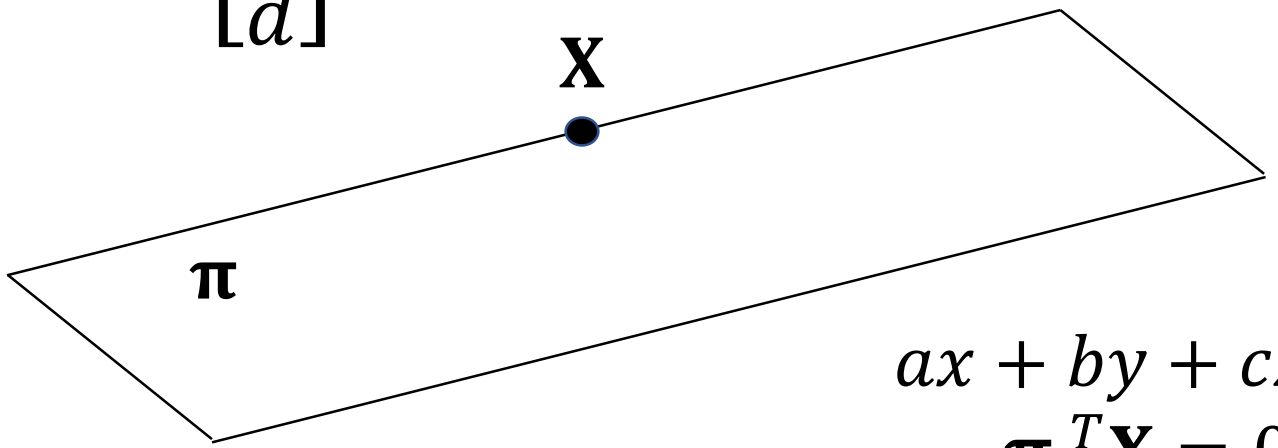
$$\boldsymbol{\pi}^T\mathbf{X} = 0 = \mathbf{X}^T\boldsymbol{\pi}$$

Incidence relation

the point $\mathbf{X} = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$ is on the plane $\boldsymbol{\pi} = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$

or

the plane $\boldsymbol{\pi} = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$ goes through the point $\mathbf{X} = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$



$$ax + by + cz + dw = 0$$

$$\boldsymbol{\pi}^T\mathbf{X} = 0 = \mathbf{X}^T\boldsymbol{\pi}$$

Dividing by $\boldsymbol{w}$ we find the cartesian coordinates again

The plane at the infinity:
the locus of the points at the infinity

The «locus» of the points at the infinity

As there are infinite points at the infinity (one for each direction), what is the aspect of the set of these points at the infinity?

Simply: $w = 0$

This set is a plane: $[\boldsymbol{a} \quad \boldsymbol{b} \quad \boldsymbol{c} \quad \boldsymbol{d}] \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \\ \boldsymbol{z} \\ \boldsymbol{w} \end{bmatrix} = 0$, actually $[\mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{1}] \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \\ \boldsymbol{z} \\ \boldsymbol{w} \end{bmatrix} = 0$

namely, **the plane at the infinity** $\boldsymbol{\pi}_\infty = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ or any $\lambda \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

NOTE: this plane has undefined normal direction

The duality principle between points and planes

Since dot product is commutative
→ incidence relation is commutative

$$\boldsymbol{\pi}^T\mathbf{X} = \begin{bmatrix} \boldsymbol{a} & \boldsymbol{b} & \boldsymbol{c} & \boldsymbol{d} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{y} \\ \boldsymbol{z} \\ \boldsymbol{w} \end{bmatrix} = 0 = \begin{bmatrix} \boldsymbol{x} & \boldsymbol{y} & \boldsymbol{z} & \boldsymbol{w} \end{bmatrix} \begin{bmatrix} \boldsymbol{a} \\ \boldsymbol{b} \\ \boldsymbol{c} \\ \boldsymbol{d} \end{bmatrix} = 0 = \mathbf{X}^T\boldsymbol{\pi} = \mathbf{0}$$

point $\mathbf{X}$ is on plane $\boldsymbol{\pi}$

$\Leftrightarrow$

point $\boldsymbol{\pi}$ is on **plane $\mathbf{X}$**

point $\mathbf{X}$ is on plane $\boldsymbol{\pi}$ (i.e. plane $\boldsymbol{\pi}$ goes through point $\mathbf{X}$)

$\leftrightarrow$

point $\boldsymbol{\pi}$ is on **plane** $\mathbf{X}$ (i.e. plane $\mathbf{X}$ goes through point $\boldsymbol{\pi}$)

Principle of **duality** between points and planes
in 3D Projective Geometry

For any true sentence containing the words

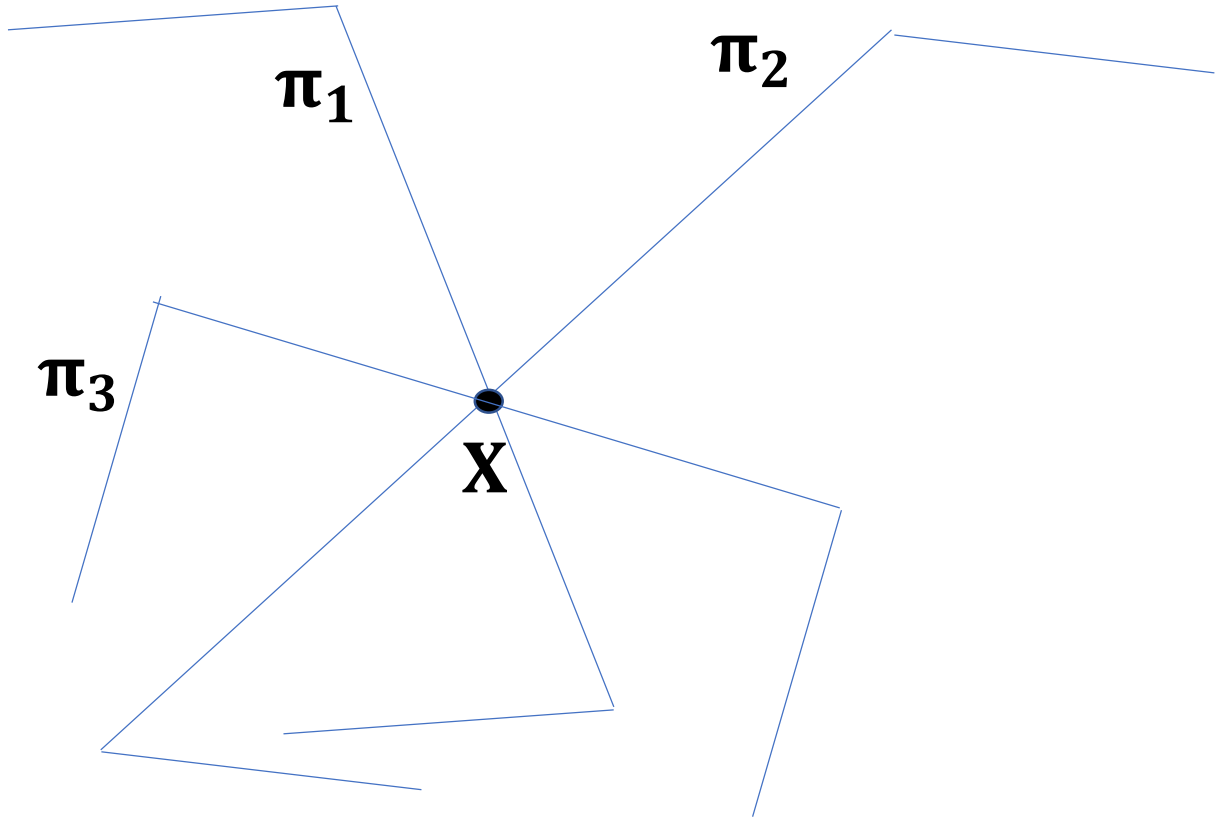
- point
- plane
- is on
- goes through

there is a DUAL sentence -also true- obtained by substituting, in the previous one, each occurence of

- point	by	- plane
- plane	by	- point
- is on	by	- goes through
- goes through	by	- is on

The point on three planes

The point on three planes



$$\begin{cases} \boldsymbol{\pi_1}^T\mathbf{X} = 0 \\ \boldsymbol{\pi_2}^T\mathbf{X} = 0 \\ \boldsymbol{\pi_3}^T\mathbf{X} = 0 \end{cases}$$

$$\begin{bmatrix} \boldsymbol{\pi_1}^T \\ \boldsymbol{\pi_2}^T \\ \boldsymbol{\pi_3}^T \end{bmatrix}_{3 \times 4} \mathbf{X} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

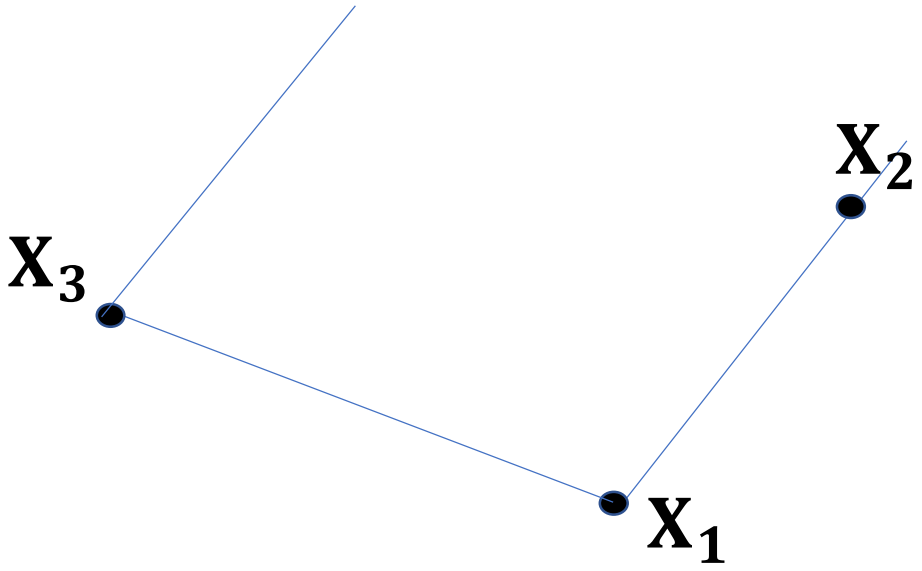
$$\mathbf{X} = \mathbf{RNS}\left(\begin{bmatrix} \boldsymbol{\pi_1}^T \\ \boldsymbol{\pi_2}^T \\ \boldsymbol{\pi_3}^T \end{bmatrix}_{3 \times 4}\right)$$

a solution vector + all its mutiples

The plane through three points

The plane through three points: DUAL of the point on three planes

The plane through three points



$$\begin{cases} \mathbf{X_1}^T\boldsymbol{\pi} = 0 \\ \mathbf{X_2}^T\boldsymbol{\pi} = 0 \\ \mathbf{X_3}^T\boldsymbol{\pi} = 0 \end{cases}$$

$$\begin{bmatrix} \mathbf{X_1}^T \\ \mathbf{X_2}^T \\ \mathbf{X_3}^T \end{bmatrix}_{3 \times 4} \boldsymbol{\pi} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\boldsymbol{\pi} = \mathbf{RNS}\left(\begin{bmatrix} \mathbf{X_1}^T \\ \mathbf{X_2}^T \\ \mathbf{X_3}^T \end{bmatrix}_{3 \times 4}\right)$$

a solution vector + all its mutiples

Lines

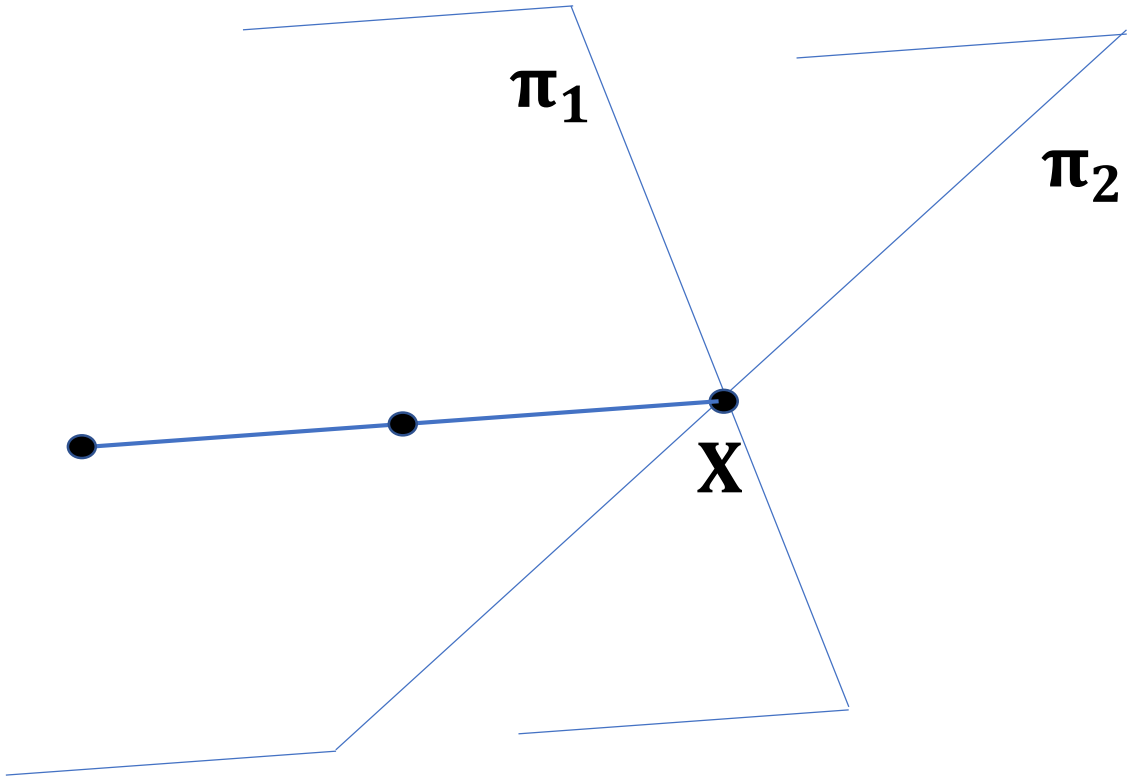
WHAT ABOUT LINES ?

Lines are primitive elements in the planar geometry
but they are **not** primitive elements in the **space** geometry

Lines can be defined through points or through planes.

Lines are intermediate entities between points and planes
they are self-dual

Line: the set of points $\mathbf{X}$ on **two** planes



$$\begin{cases} \boldsymbol{\pi_1}^T \mathbf{X} = 0 \\ \boldsymbol{\pi_2}^T \mathbf{X} = 0 \end{cases}$$

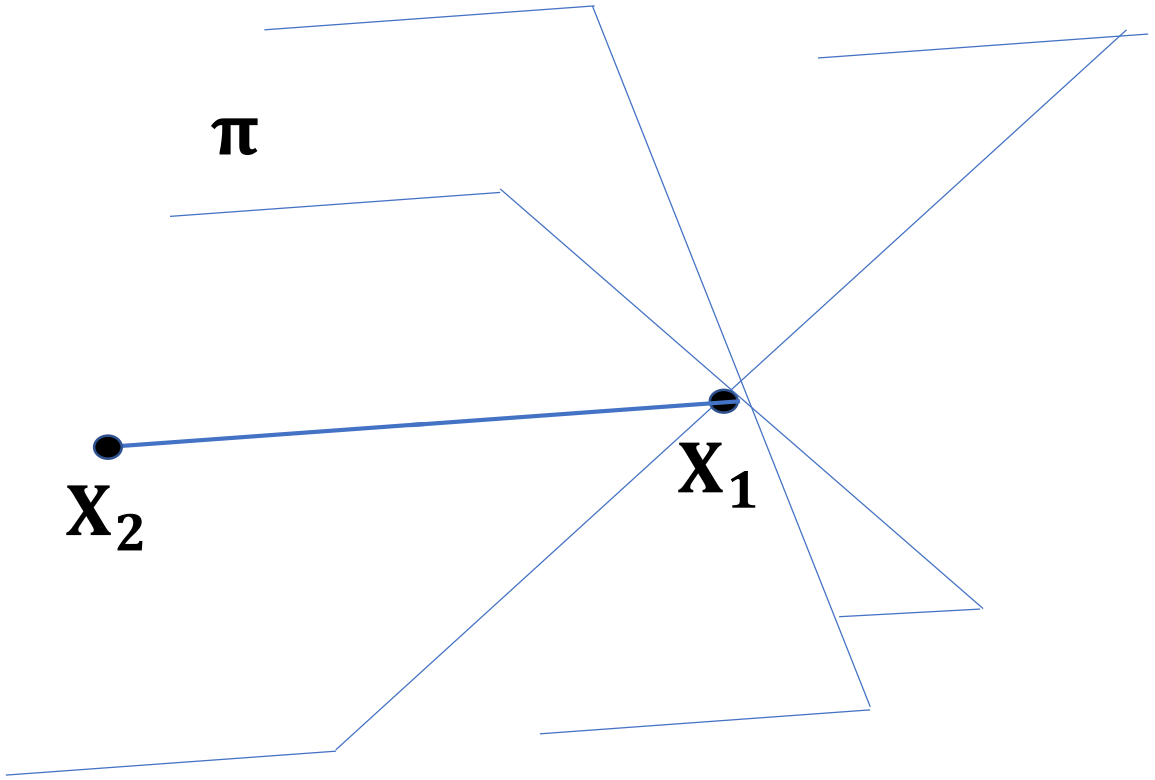
$$\begin{bmatrix} \boldsymbol{\pi_1}^T \\ \boldsymbol{\pi_2}^T \end{bmatrix}_{2 \times 4} \mathbf{X} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\mathbf{L} = \mathbf{RNS}\left(\begin{bmatrix} \boldsymbol{\pi_1}^T \\ \boldsymbol{\pi_2}^T \end{bmatrix}_{2 \times 4}\right)$$

2D set of solution vectors: two points and all their linear combinations

→ due to homogeneity: 1D set of points (parameter abscissa)

Line: the set of planes **π** through **two** points



$$\begin{cases} \mathbf{X_1}^T \boldsymbol{\pi} = 0 \\ \mathbf{X_2}^T \boldsymbol{\pi} = 0 \end{cases}$$

$$\begin{bmatrix} \mathbf{X_1}^T \\ \mathbf{X_2}^T \end{bmatrix}_{2 \times 4} \boldsymbol{\pi} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

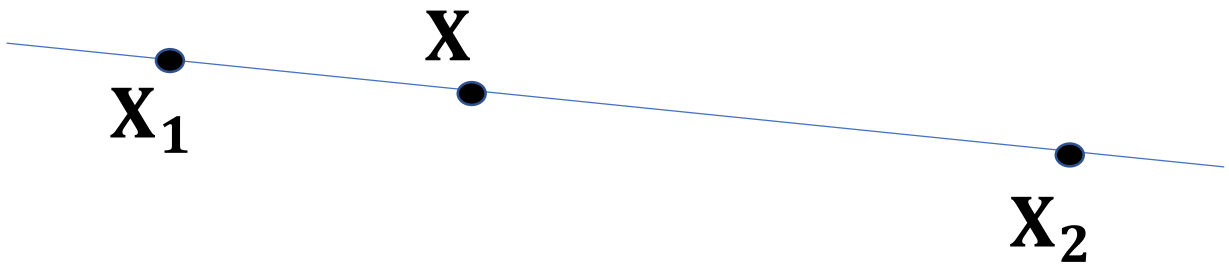
$$\mathbf{L}^* = \mathbf{RNS}\left(\begin{bmatrix} \mathbf{X_1}^T \\ \mathbf{X_2}^T \end{bmatrix}_{2 \times 4}\right)$$

2D set of vector solutions: two planes and all their linear combinations

→ due to homogeneity: 1D set of planes (parameter: rotation angle)

linear combination of two points

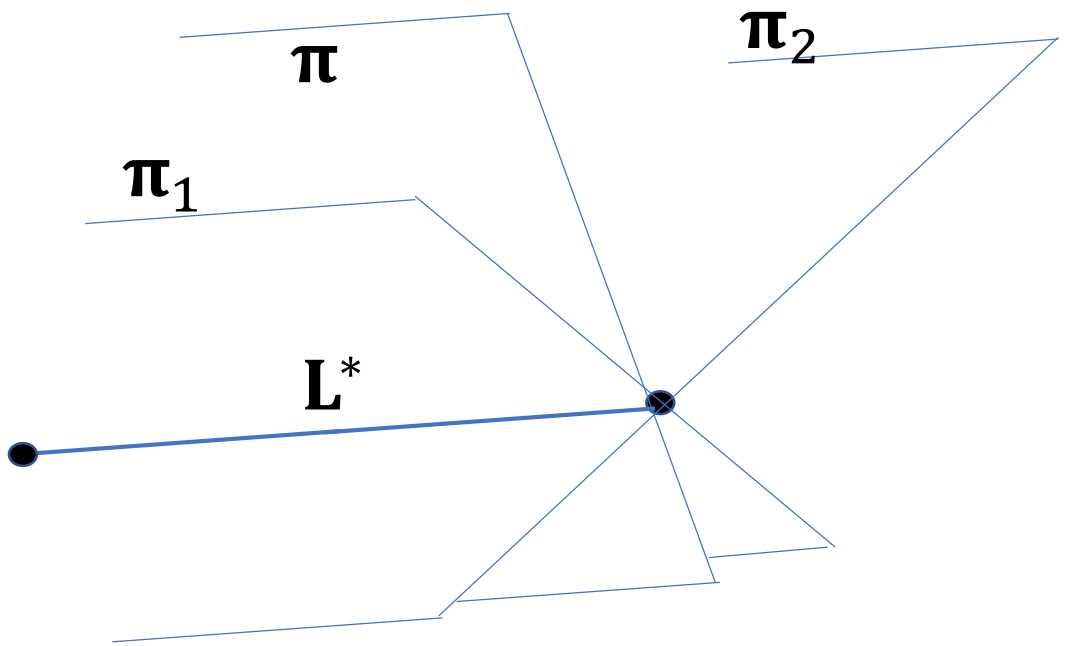
Property: the point $\mathbf{X}$ given by the linear combination $\mathbf{X} = \propto \mathbf{X}_1 + \beta \mathbf{X}_2$ of two points $\mathbf{X}_1$ and $\mathbf{X}_2$ is on the line $\mathbf{L}$ through $\mathbf{X}_1$ and $\mathbf{X}_2$



A **line** $\mathbf{L}$ can also be defined as the set of all points, that are linear combinations of two given points: $\mathbf{X} = \propto \mathbf{X}_1 + \beta \mathbf{X}_2$

DUAL: linear combination of two planes

Dual property: the plane $\boldsymbol{\pi}$**,** given by the linear combination $\boldsymbol{\pi} = \propto \boldsymbol{\pi}_1 + \beta\boldsymbol{\pi}_2$ of two planes $\boldsymbol{\pi}_1$ and $\boldsymbol{\pi}_2$, goes through the line $\mathbf{L}^*$ on $\boldsymbol{\pi}_1$ and $\boldsymbol{\pi}_2$

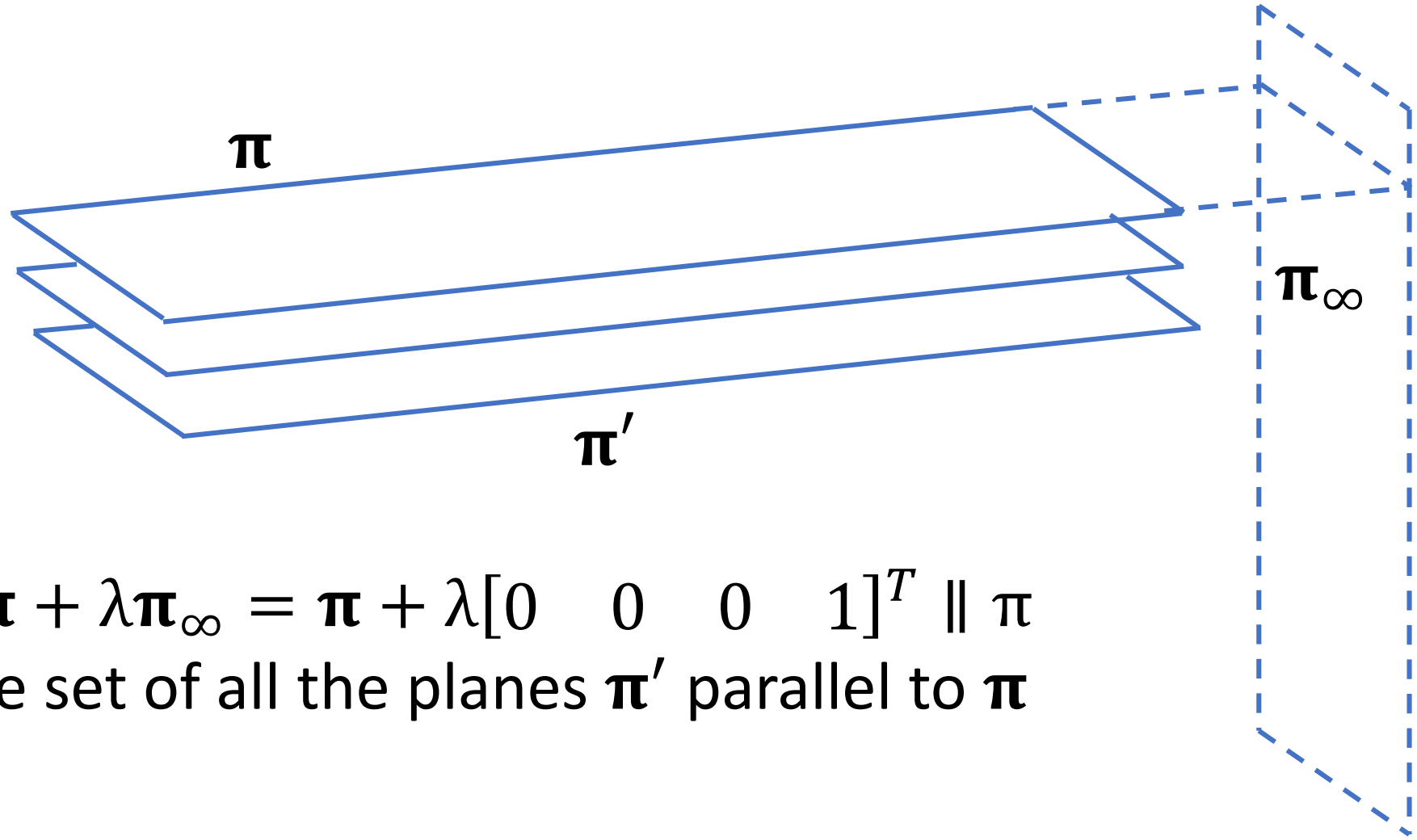


A **line** $\mathbf{L}^*$ can also be defined as the set of all planes, that are linear combinations of two given planes: $\boldsymbol{\pi} = \propto \boldsymbol{\pi}_1 + \beta\ \boldsymbol{\pi}_2$

pairs of DUALLY corresponding words

- point	→	- plane
- line	→	- line
- plane	→	- point
- is on	→	- goes through
- goes through	→	- is on

A special case:
linear combinations of a plane $\boldsymbol{\pi}$ and the plane $\boldsymbol{\pi}_\infty$



$$\boldsymbol{\pi}' = \boldsymbol{\pi} + \lambda\boldsymbol{\pi}_\infty = \boldsymbol{\pi} + \lambda[0 \quad 0 \quad 0 \quad 1]^T \parallel \pi$$

→ the set of all the planes $\boldsymbol{\pi}'$ parallel to $\boldsymbol{\pi}$

Remark: planes and lines at the infinity

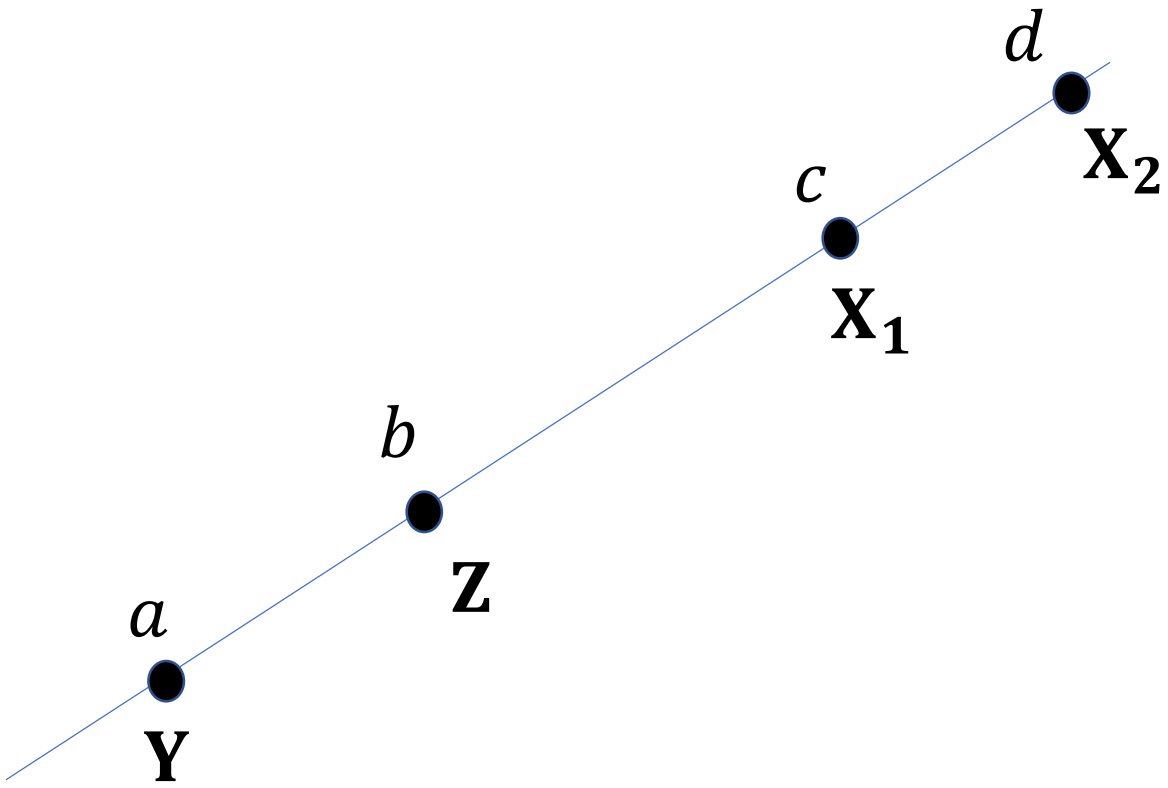
Each plane $\boldsymbol{\pi}$ has its own line at the infinity $\boldsymbol{l_\infty(\pi)}$
and also its own circular points $\boldsymbol{I(\pi)}$ and $\boldsymbol{J(\pi)}$

parallel planes share the same $\boldsymbol{l_\infty}$
and the same circular points $\boldsymbol{I}$ and $\boldsymbol{J}$

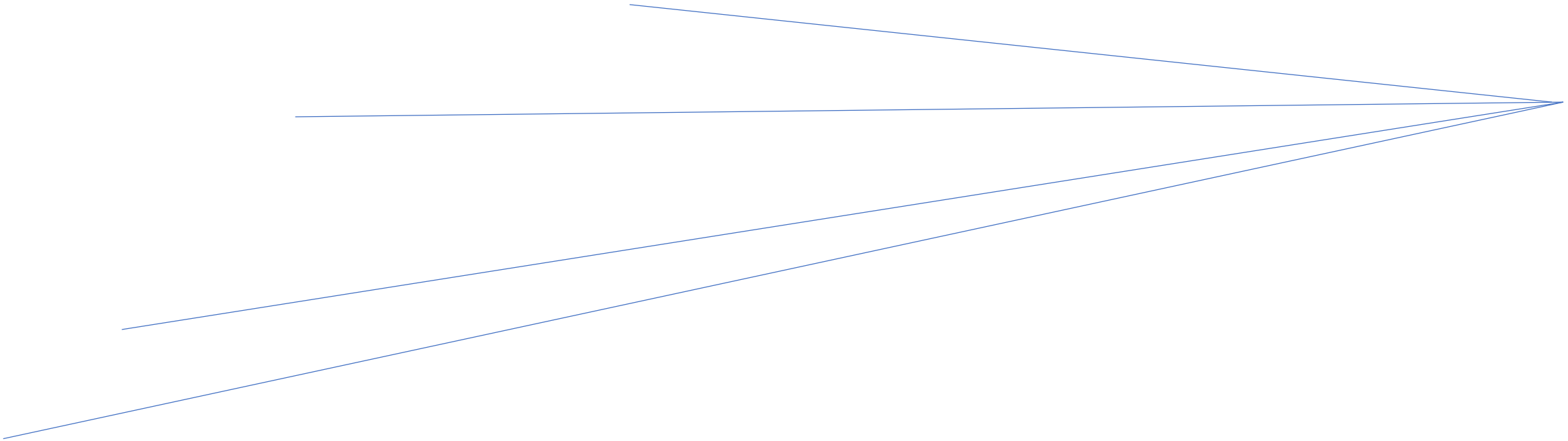
The cross ratio in 2D and in 3D

1D cross ratio of a 4-tuple of colinear points

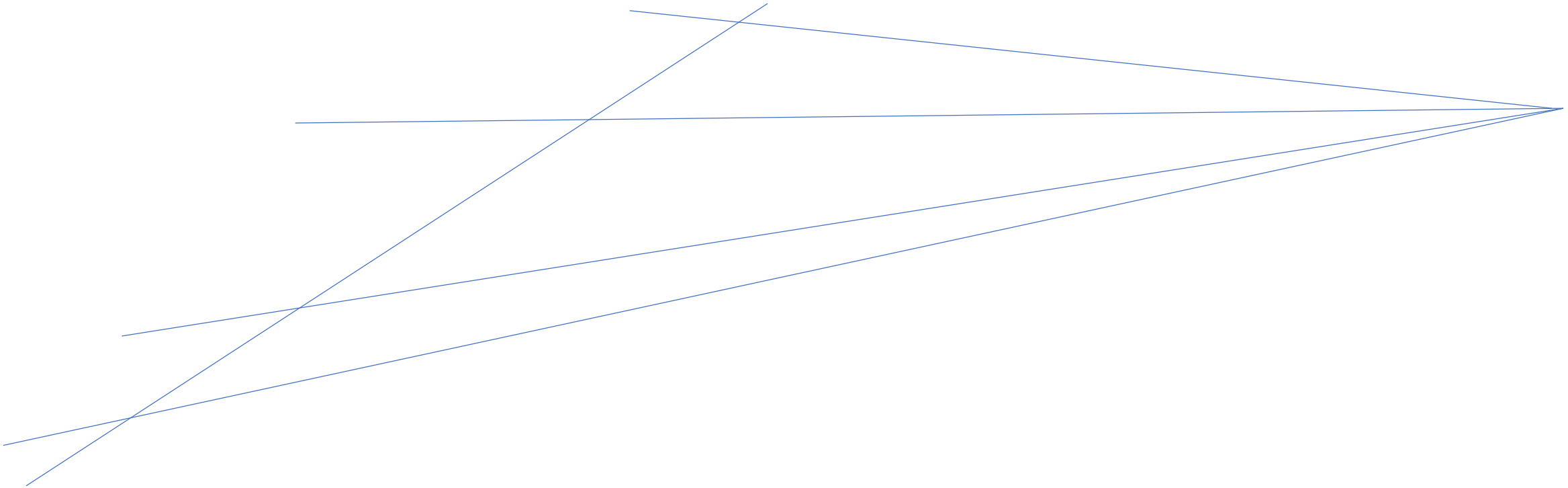
$$CR_{Y,Z,\mathrm{X}_1,\mathrm{X}_2} = \frac{c-a}{c-b} \Big/ \frac{d-a}{d-b}$$



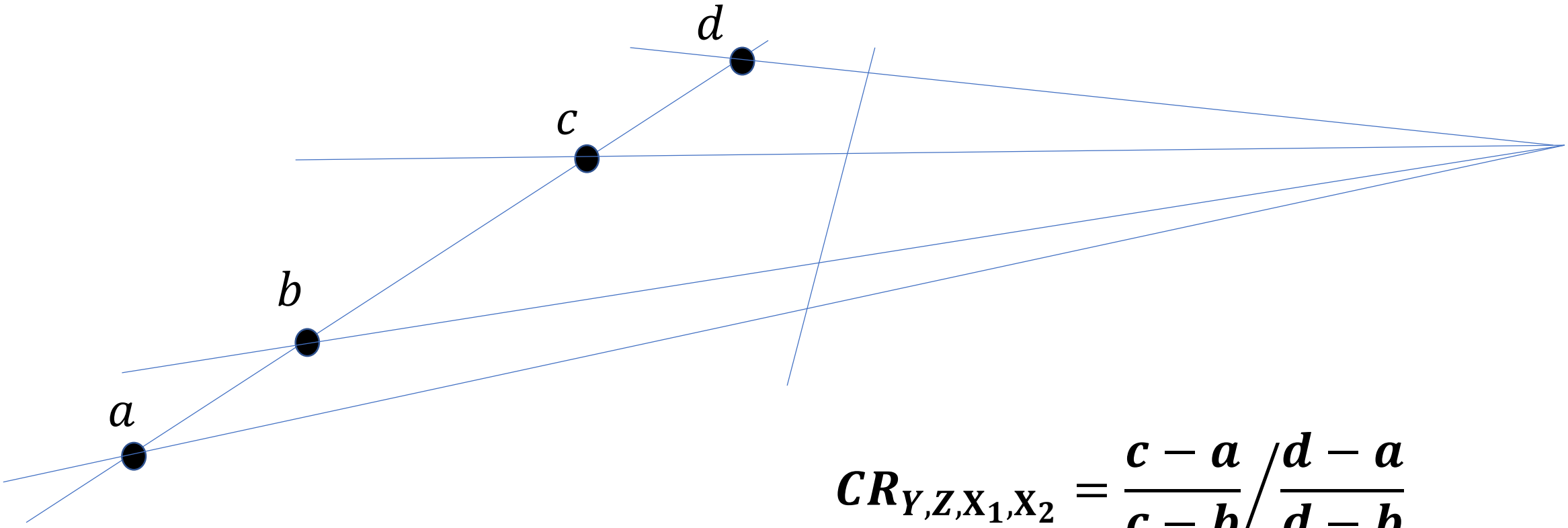
2D cross ratio of a 4-tuple of coplanar, concurrent lines



2D cross ratio of a 4-tuple of coplanar, concurrent lines: take any crossing line …



2D cross ratio of a 4-tuple of coplanar, concurrent lines: take any crossing line … compute the 1D cross ratio of intersection points

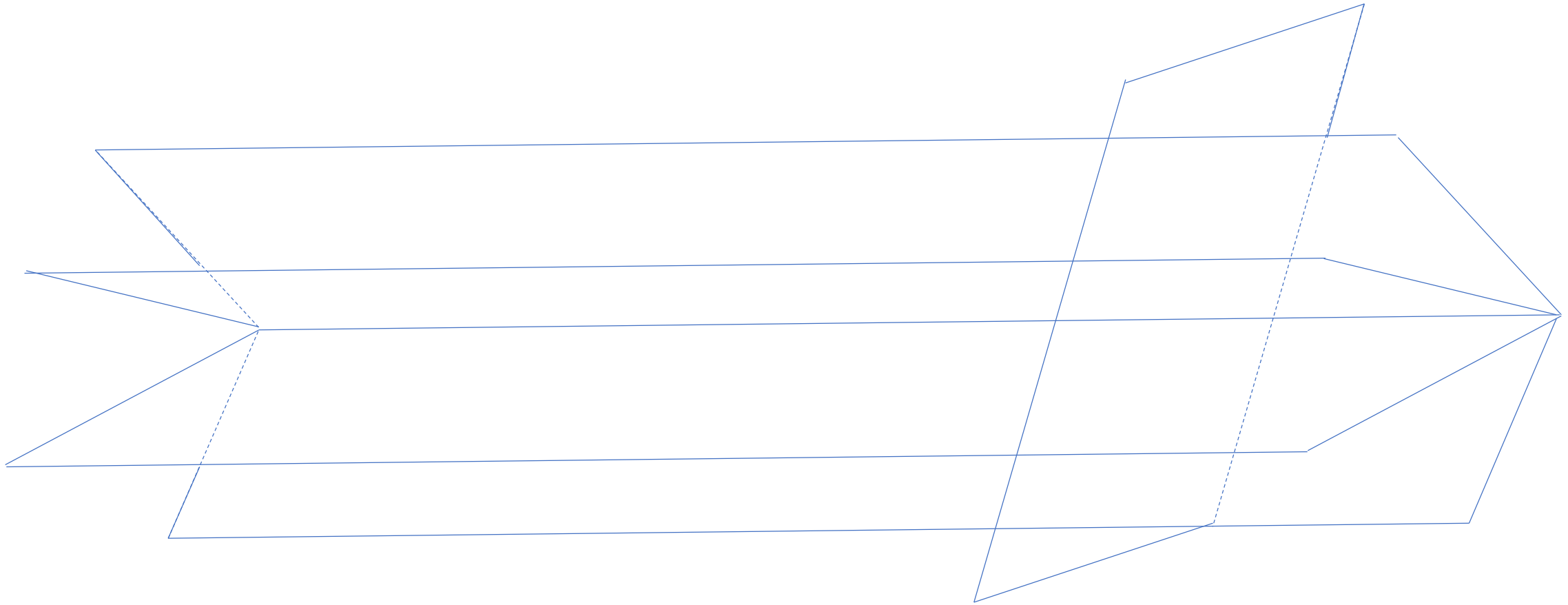


$$\boldsymbol{CR_{Y,Z,X_1,X_2}} = \frac{\boldsymbol{c-a}}{\boldsymbol{c-b}} \Big/ \frac{\boldsymbol{d-a}}{\boldsymbol{d-b}}$$

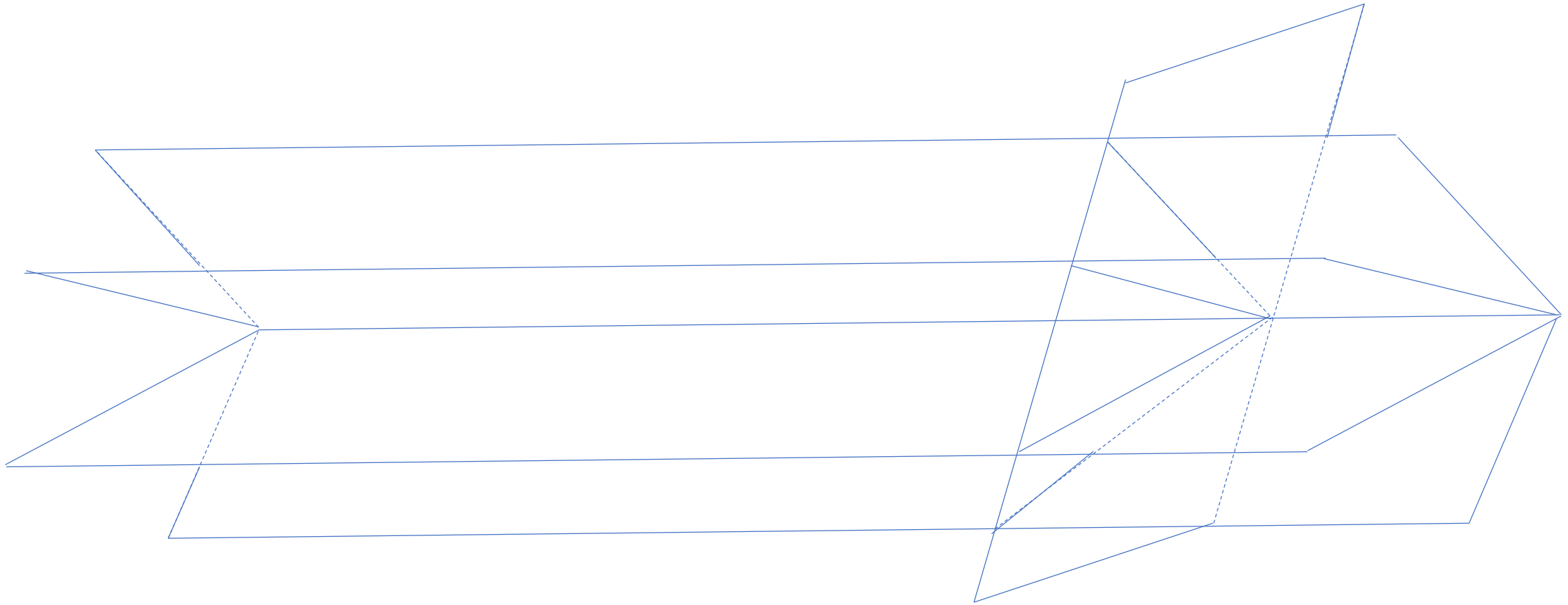
3D cross ratio of a 4-tuple of coaxial planes:



3D cross ratio of a 4-tuple of coaxial planes: take any crossing plane …

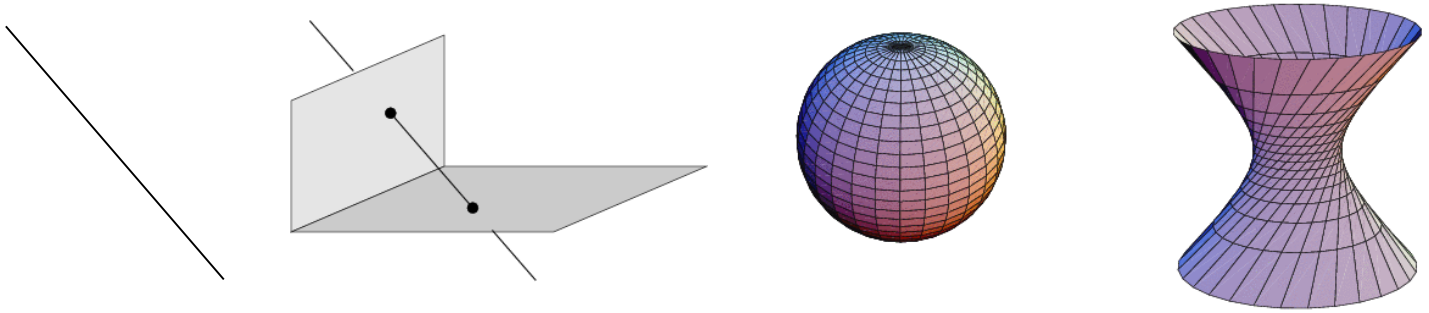


3D cross ratio of a 4-tuple of coaxial planes: take any crossing plane … compute the 2D cross ratio of intersection lines

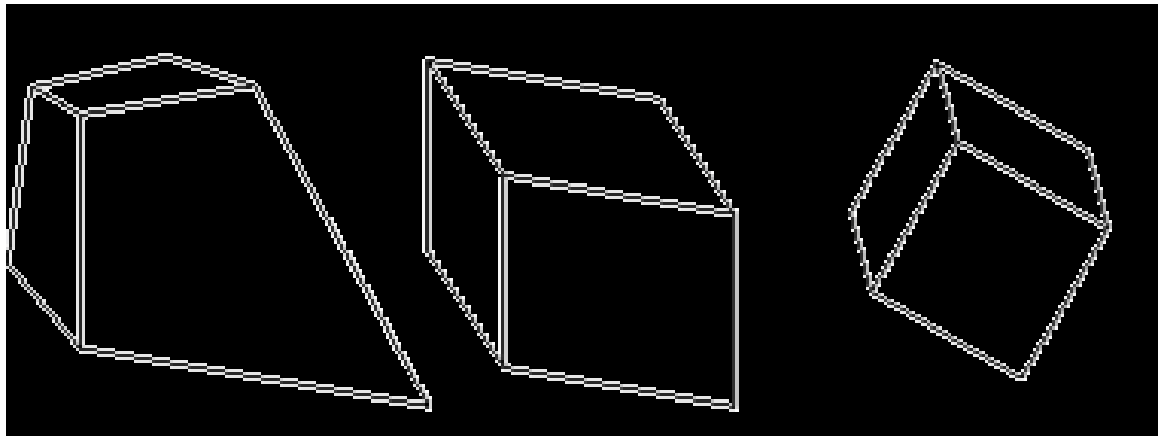


3D Space Projective Geometry

- **Elements**
- Points
- Planes
- **Quadrics**
- (Dual quadrics)



- **Transformations**
- Isometries
- Similarities
- Affinities
- Projectivities



Quadrics

Quadric: a point $\mathbf{X}$ is on a quadric $\boldsymbol{Q}$ if it satisfies a homogeneous *quadratic* equation, namely

$$\mathbf{X}^T \boldsymbol{Q}\, \mathbf{X} = \mathbf{0}$$

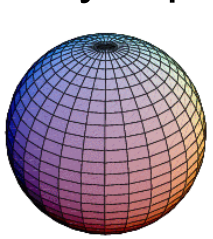
where $\boldsymbol{Q}$ is a 4x4 symmetric matrix.

$$\mathrm{Q} = \begin{bmatrix} \bullet & \bullet & \bullet & \bullet \\ \circ & \bullet & \bullet & \bullet \\ \circ & \circ & \bullet & \bullet \\ \circ & \circ & \circ & \bullet \end{bmatrix}$$

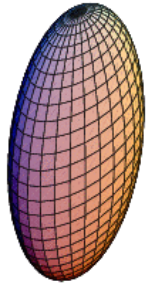
- $\boldsymbol{Q}$ is a homogeneous matrix: $\lambda\boldsymbol{Q} \Leftrightarrow \boldsymbol{Q}$
- 9 degrees of freedom
- 9 points in general positions define a quadric

Quadric classification

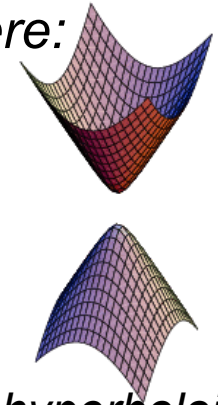
Projectively equivalent to *sphere:*



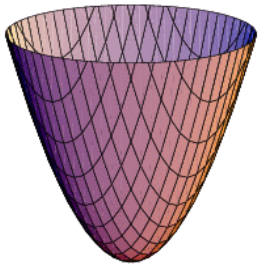
sphere



ellipsoid

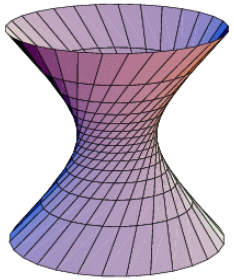


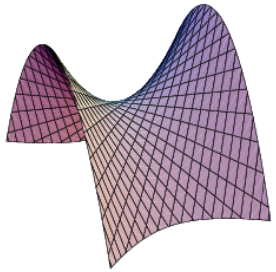
hyperboloid of two sheets



paraboloid

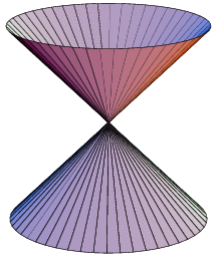
Ruled quadrics:



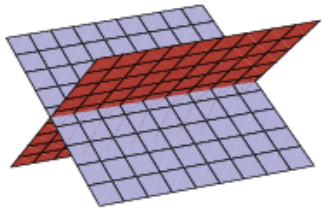


hyperboloids of one sheet

Degenerate ruled quadrics:



cone



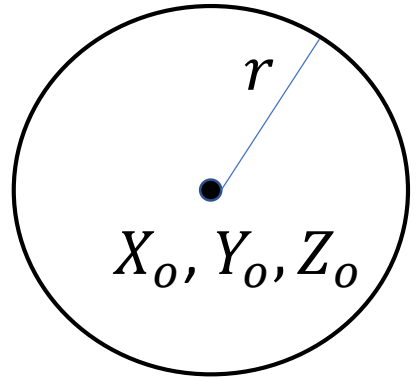
two planes

Example: the sphere

First in cartesian coordinates:

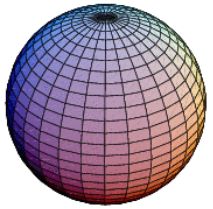
$$(X - X_o)^2 + (Y - Y_o)^2 + (Z - Z_o)^2 - r^2 = 0$$

X_o, Y_o, Z_o are the center coordinates, r is the radius.



… then in homogeneous coordinates:

$$[x \quad y \quad z \quad w] \begin{bmatrix} 1 & 0 & 0 & -X_o \\ 0 & 1 & 0 & -Y_o \\ 0 & 0 & 1 & -Z_o \\ -X_o & -Y_o & -Z_o & X_o^2 + Y_o^2 - r^2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = 0$$



$$\mathbf{X}^\intercal \; \boldsymbol{Q} \; \mathbf{X} = 0$$

Digression:
Intersection of a plane and a quadric

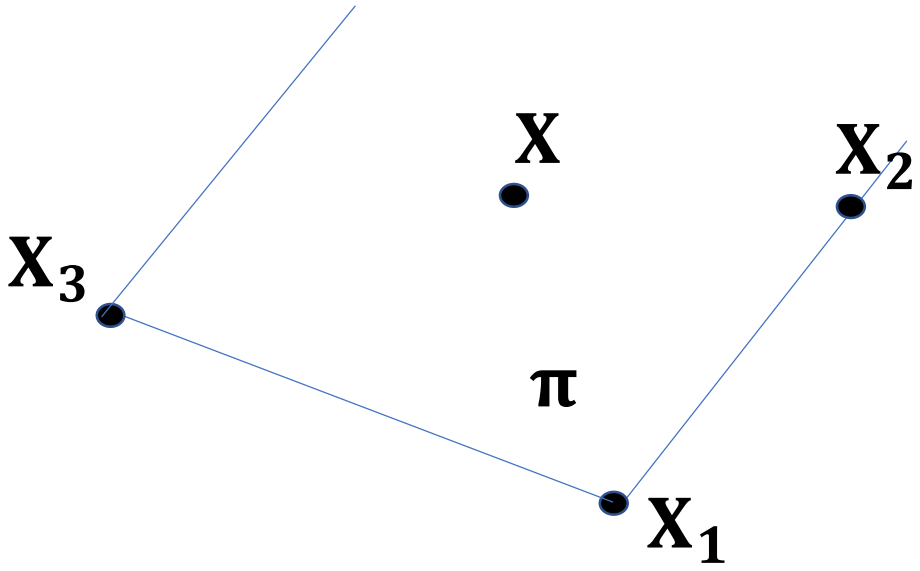
A plane can be defined by its **span**:

$$\boldsymbol{\pi}: \mathbf{X} = \mathbf{M}\mathbf{x} \qquad (\mathbf{M} \text{ is a 4x3 matrix})$$

where vector $\mathbf{x}$ represent homogeneous coordinates within $\boldsymbol{\pi}$

…

Defining a plane by its **span**



$\mathbf{X}$ is a linear combination $\propto \mathbf{X_1} + \beta\mathbf{X_2} + \gamma\mathbf{X_3}$

→ $\mathbf{X}$ is coplanar to $\mathbf{X_1}, \mathbf{X_2},$ and $\mathbf{X_3}$

i.e. $\mathbf{X} = [\mathbf{X_1} \quad \mathbf{X_2} \quad \mathbf{X_3}] \begin{bmatrix} \propto \\ \beta \\ \gamma \end{bmatrix} = \mathrm{M}\mathbf{x}$ where

$\mathbf{x} = \begin{bmatrix} \propto \\ \beta \\ \gamma \end{bmatrix}$ can be regarded as homogeneous coordinates within the 2D geometry of plane $\boldsymbol{\pi}$

$$\mathbf{X} = \mathrm{M}\mathbf{x}$$

Intersection of a plane and a quadric

Plane defined by its span: $\boldsymbol{\pi}: \mathbf{X} = \mathbf{M}\mathbf{x}$ ($\mathbf{M}$ is a 4x3 matrix)
where vector $\mathbf{x}$ represent homogeneous coordinates within $\boldsymbol{\pi}$

Quadric $\mathbf{Q}$: $\mathbf{X}^T \boldsymbol{Q}\, \mathbf{X} = \mathbf{0}$

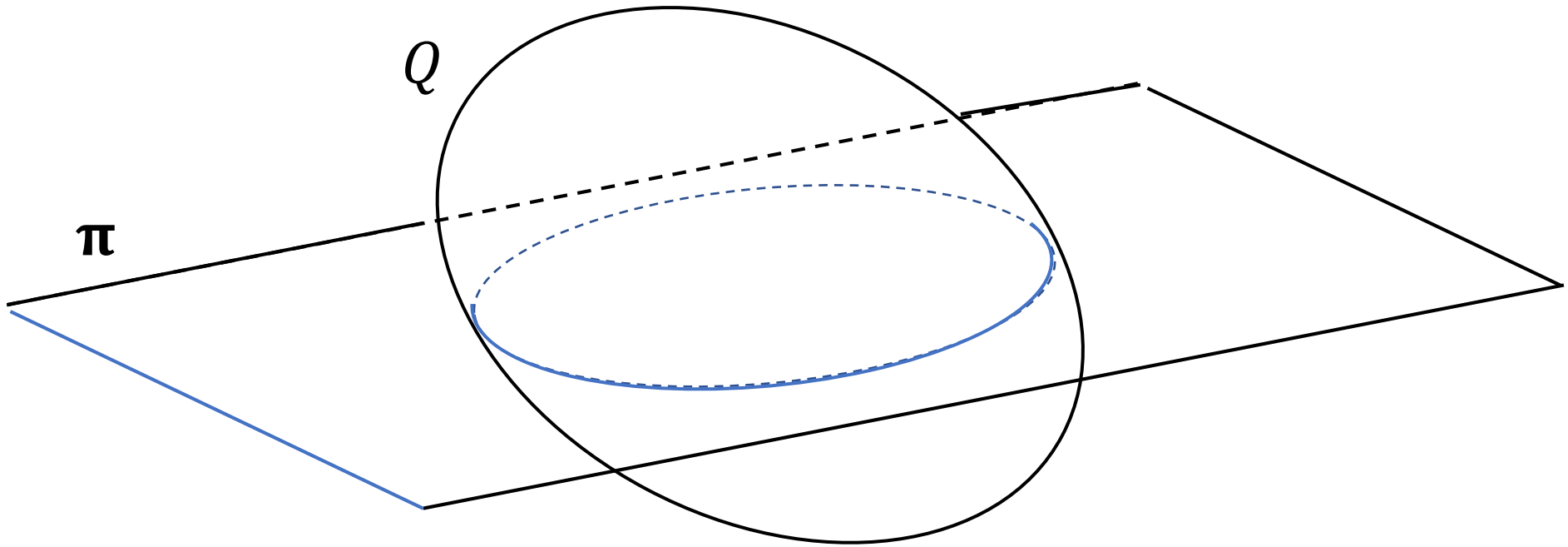
Plane-quadric intersection: $\mathbf{X}^T \boldsymbol{Q}\, \mathbf{X} = \mathbf{x}^T \mathbf{M}^T \boldsymbol{Q} \mathbf{M}\mathbf{x} = \mathbf{x}^T \boldsymbol{C}\, \mathbf{x} = \mathbf{0}$

… is a **conic** $\boldsymbol{C}$: a set of points $\mathbf{x}$ within the plane $\boldsymbol{\pi}$

$$\boldsymbol{C} = \mathbf{M}^T \boldsymbol{Q} \mathbf{M}$$

plane – quadric intersection: a conic

.



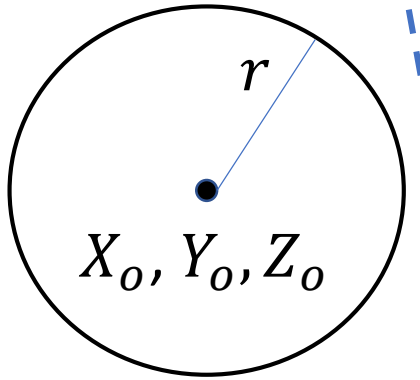
The absolute conic:
an extension of the circular points

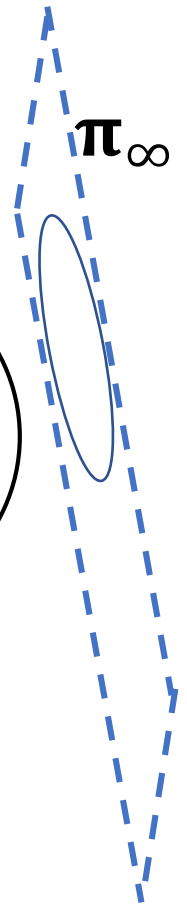
A noteworthy example: intersecting a sphere and the plane at the ∞

$$\begin{cases} (x - X_o w)^2 + (y - Y_o w)^2 + (z - Z_o w)^2 - r^2 w^2 = 0 \\ w = 0 \end{cases}$$

→

$$\begin{cases} x^2 + y^2 + z^2 = 0 \\ w = 0 \end{cases}$$



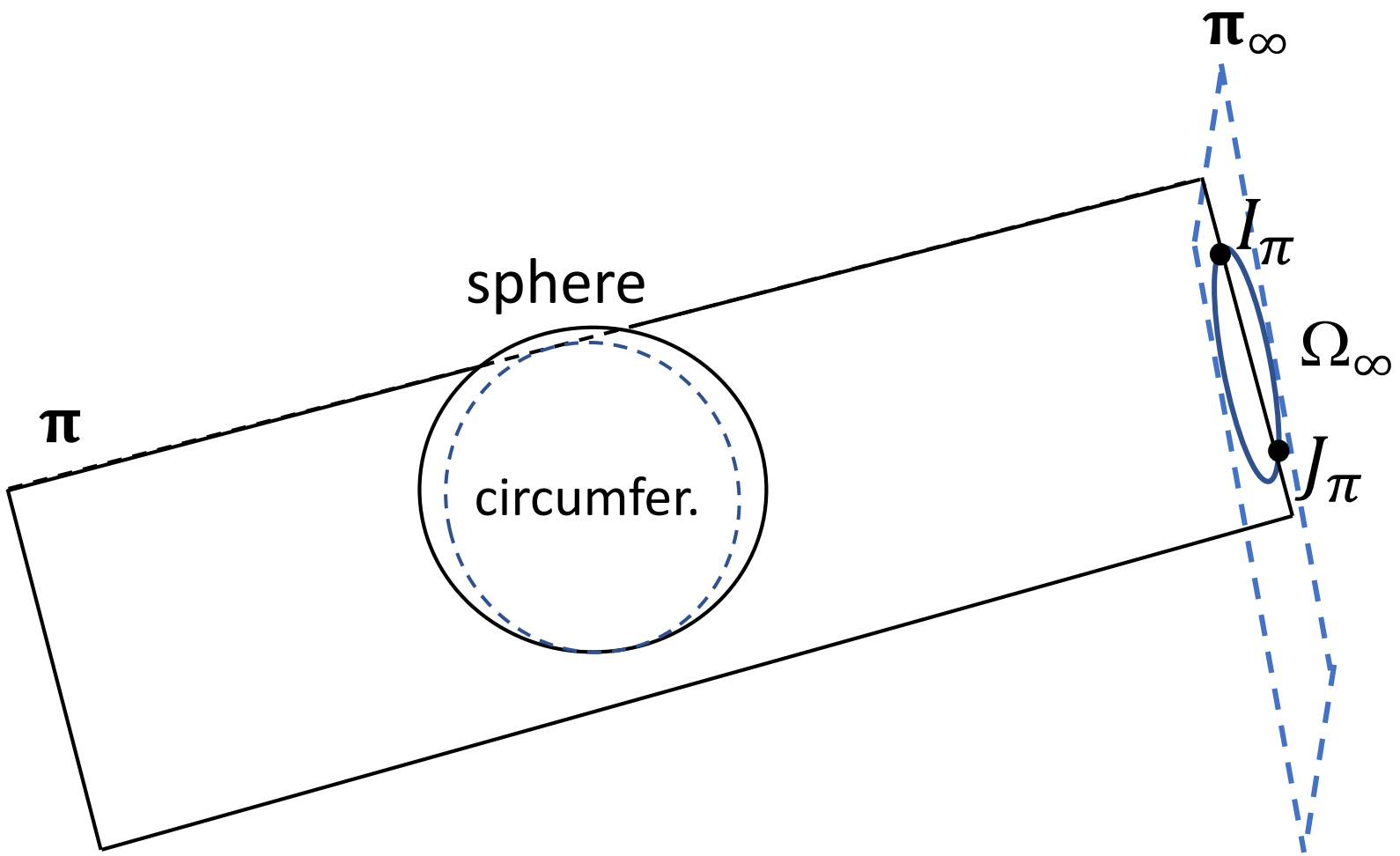


The sphere parameters (center and radius) disappear from the equation → the intersection **conic** is the **same for all** spheres:

$$x^2 + y^2 + z^2 = \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x & y & z \end{bmatrix} \Omega_\infty \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

A conic within $\boldsymbol{\pi}_\infty$: $\Omega_\infty = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **ABSOLUTE CONIC**

absolute conic: made of (∪) circular points of all planes

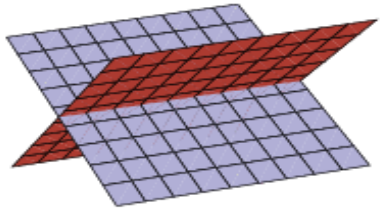


Degenerate quadrics

$$\mathbf{X}^T Q\, \mathbf{X} = \mathbf{0}$$

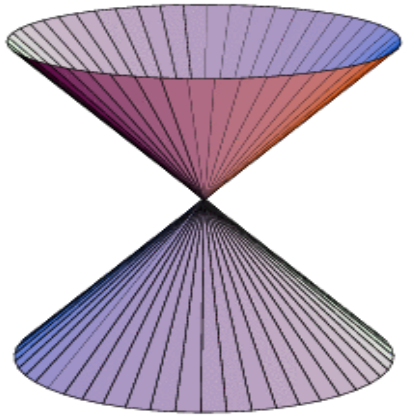
- rank $Q = 1 \rightarrow Q = \mathbf{A}\mathbf{A}^T$ repeated plane $\mathbf{A}$

- rank $Q = 2 \rightarrow Q = \mathbf{A}\mathbf{B}^T + \mathbf{B}\mathbf{A}^T$



two planes $\mathbf{A}$ and $\mathbf{B}$

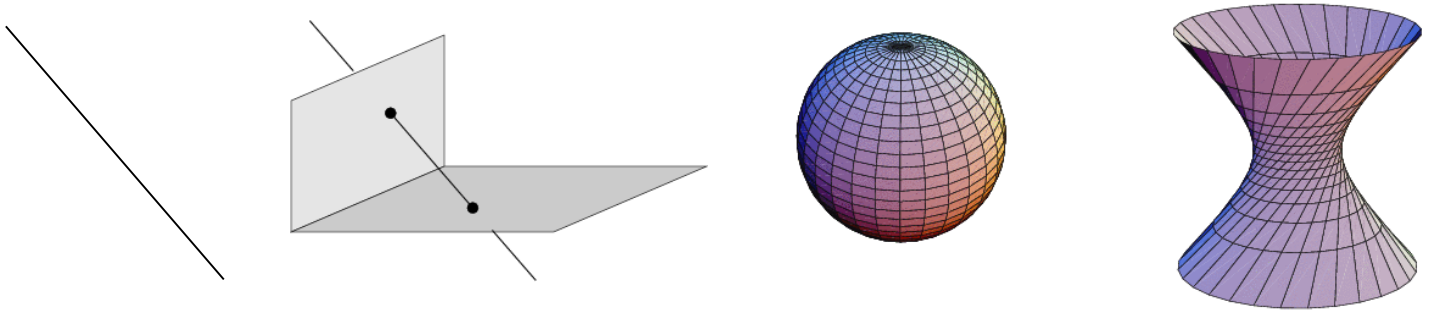
- rank $Q = 3$ a **cone**
  vertex = RNS(Q)



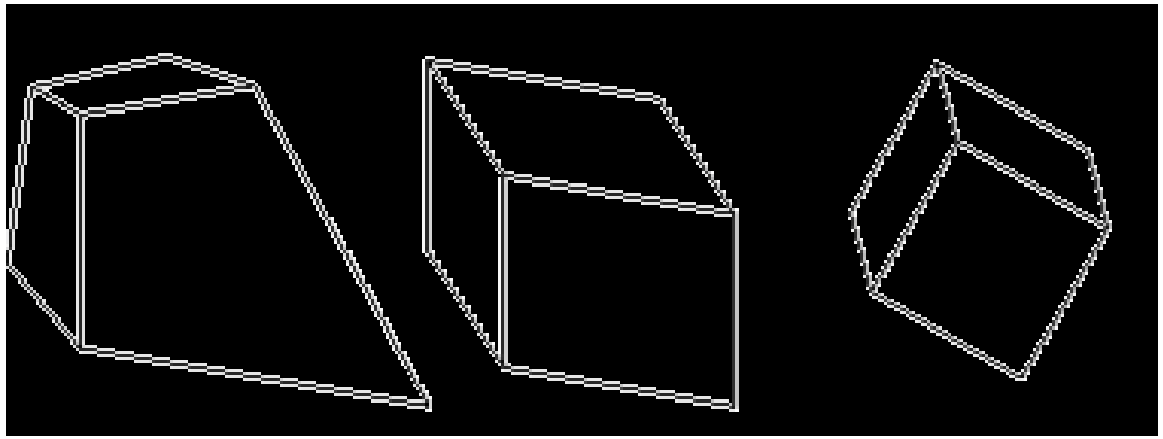
a cylinder is a cone
whose vertex is at the infinity

3D Space Projective Geometry

- **Elements**
- Points
- Planes
- Quadrics
- (Dual quadrics)



- **Transformations**
- Isometries
- Similarities
- Affinities
- Projectivities



Projective 3D Geometry: Projective Transformations

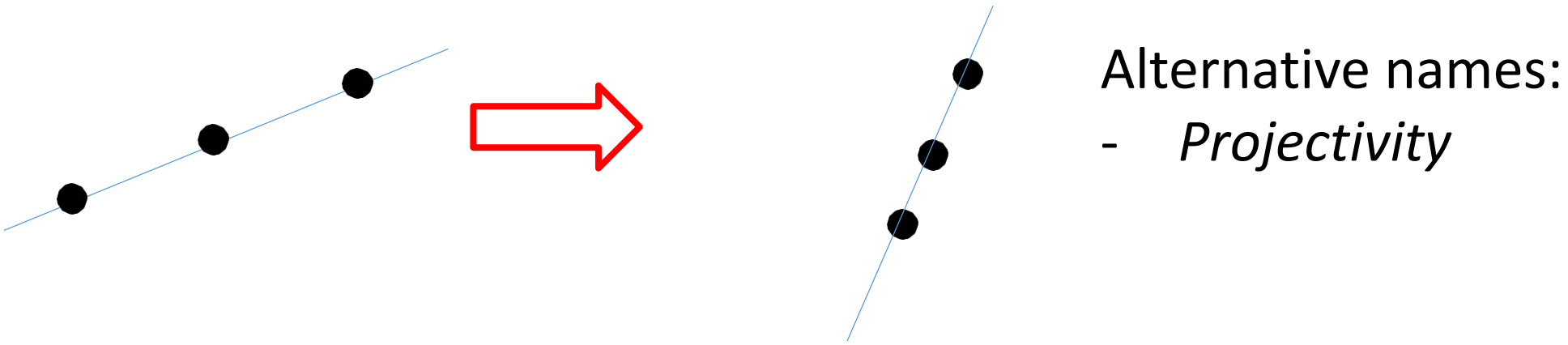
Projective mappings

Def. *A **projective mapping** between a projective space* $\mathbb{P}^3$ *and an other projective space* $\mathbb{P}'^3$ *is an **invertible** mapping which preserves colinearity:*

$$h\colon \mathbb{P}^3 \rightarrow \mathbb{P}'^3, \mathbf{X}' = h(\mathbf{X}),\ \mathbf{X}_1,\ \mathbf{X}_2, \mathbf{X}_3 \text{ are colinear}$$

$$\leftrightarrow$$

$$\mathbf{X}'_1 = h(\mathbf{X_1}),\ \mathbf{X}'_2 = h(\mathbf{X}_2), \mathbf{X}'_3 = h(\mathbf{X}_3) \text{ are colinear}$$



Alternative names:

- *Projectivity*

Fundamental Theorem of Projective Geometry

Theorem: *A mapping* $h : \mathbb{P}^3 \rightarrow \mathbb{P}'^3$ *is projective* if and only if *there exists an invertible 4x4 matrix* H *such that for any point in* $\mathbb{P}^3$ *represented by the vector* $\mathbf{X}$ *, is* $h(\mathbf{X}) = H\,\boldsymbol{X}$

i.e. projective mappings are LINEAR in the homogeneous coordinates

(they are not linear in cartesian coordinates)

Projectivity: 15 degrees of freedom

From the theorem

$$h(\mathbf{X}) = \mathbf{X}' = H\,\mathbf{X}$$

Therefore, if we multiply the matrix H by any nonzero scalar λ, the relation is satisfied by the same points

$$\mathbf{X}' = \lambda H\,\mathbf{X}$$

Thus any nonzero multiple of the matrix H represents the same projective mapping as H.

Hence H is a homogeneous matrix: in spite of its 16 entries,

H has only 15 degrees of freedom, namely the ratios between its elements.

Transformation of points, planes, quadrics, dual quadrics

Transformation rules for the space elements

A homography transforms **each point $\boldsymbol{X}$** into a point $\boldsymbol{X}'$ such that:

$$\boldsymbol{X} \rightarrow H\boldsymbol{X} = \boldsymbol{X}'$$

A homography transforms **each plane $\boldsymbol{\pi}$** into a plane $\boldsymbol{\pi}'$ such that:

$$\boldsymbol{\pi} \rightarrow H^{-T}\boldsymbol{\pi} = \boldsymbol{\pi}'$$

A homography transforms **each quadric $\boldsymbol{Q}$** into a quadric Q′ such that:

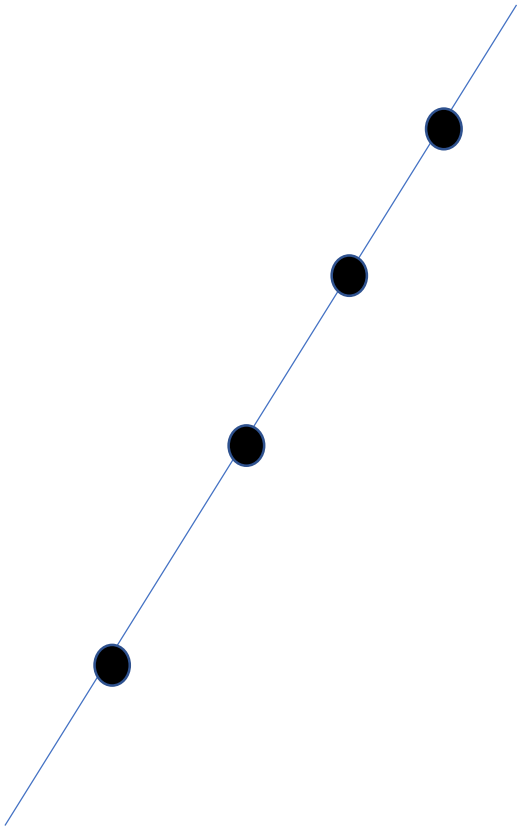
$$Q \rightarrow H^{-T}QH^{-1} = Q'$$

A homography transforms **each dual quadric** Q^* into a dual quadric $Q^{*\prime}$

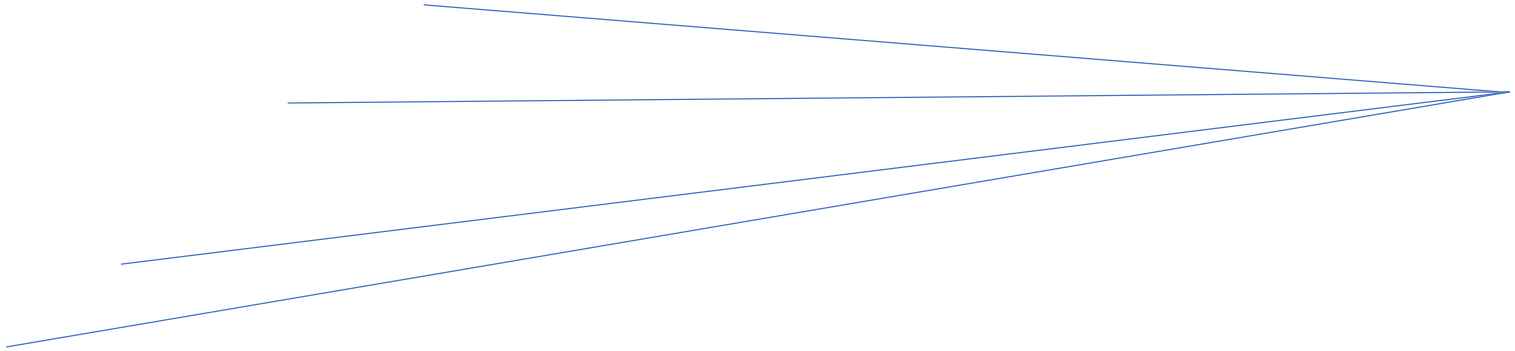
$$Q^* \rightarrow HQ^*H^T = Q^{*\prime}$$

Cross ratios: invariant under projective mappings

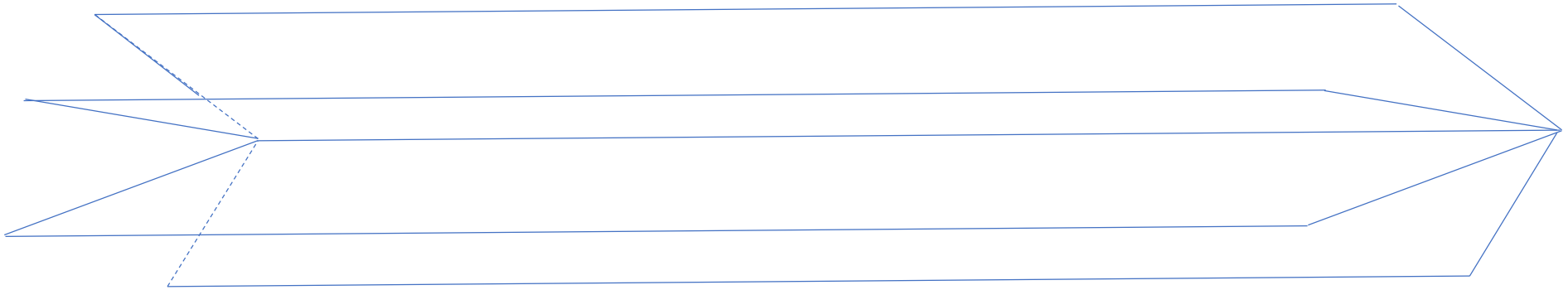
1D cross ratio of a 4-tuple of colinear points



2D cross ratio of a 4-tuple of concurrent, coplanar lines



3D cross ratio of a 4-tuple of co-axial lines

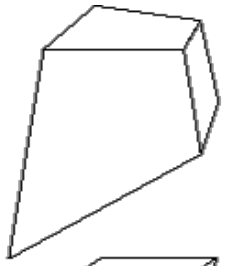


Hierarchy of projective transformations

Hierarchy of transformations

Projective
15dof

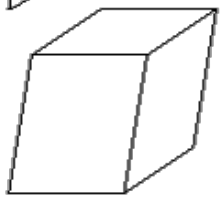
$$\begin{bmatrix} A & t \\ v^T & v \end{bmatrix}$$



Intersection and tangency

Affine
12dof

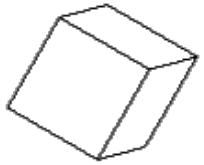
$$\begin{bmatrix} A & t \\ 0^T & 1 \end{bmatrix}$$



Parallellism of planes,
Volume ratios, centroids,
The plane at infinity π_∞

Similarity
7dof

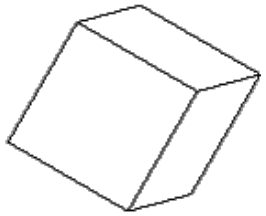
$$\begin{bmatrix} sR & t \\ 0^T & 1 \end{bmatrix}$$



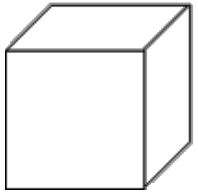
The absolute conic Ω_∞

Euclidean
6dof

$$\begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix}$$



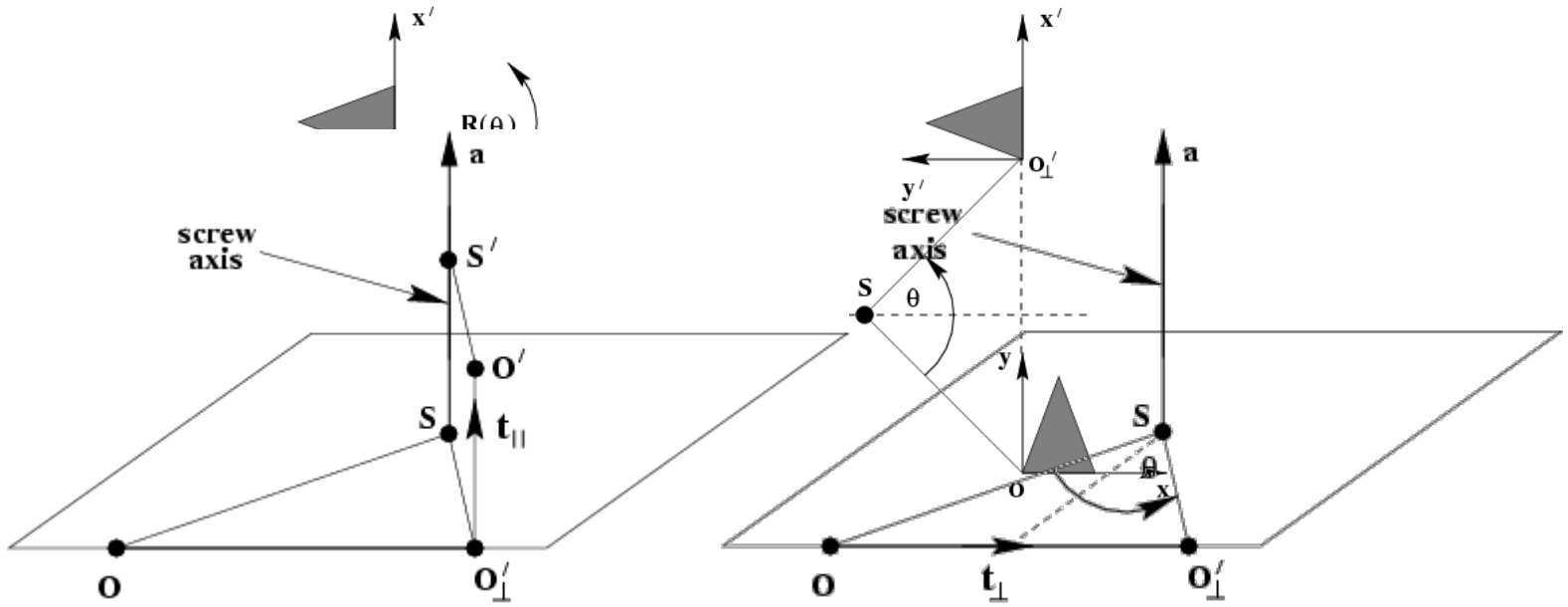
Volume



Caveat

A 3D rototranslation **is not** a pure rotation: screw decomposition

Any rotor-translation is equivalent to a rotation about a screw axis and a translation along the screw axis.



screw axis // rotation axis

$$t = t_{//} + t_{\perp}$$

Isometries (or Euclidean mappings)

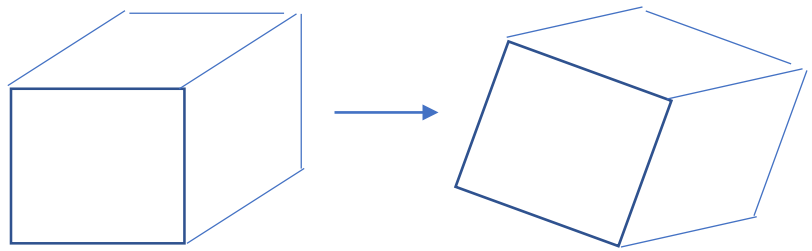
$$H_I = \begin{bmatrix} R_\perp & \boldsymbol{t} \\ 0 \quad 0 & 0 \quad 1 \end{bmatrix}$$

$R_\perp$ is a 3x3 orthogonal matrix: ${R_\perp}^{-1} = {R_\perp}^T$

$\det {R_\perp}^{-1} = 1$ planar rigid displacement (-1 for reflection)

6 dofs: translation $\boldsymbol{t}$ + Euler angles $\vartheta, \varphi,$ □

Invariants: lenghts, distances, areas → shape and size → relative positions



Similarities

$$H_S = \begin{bmatrix} s\,R_\perp & \boldsymbol{t} \\ 0 \quad 0 & 1 \end{bmatrix}$$

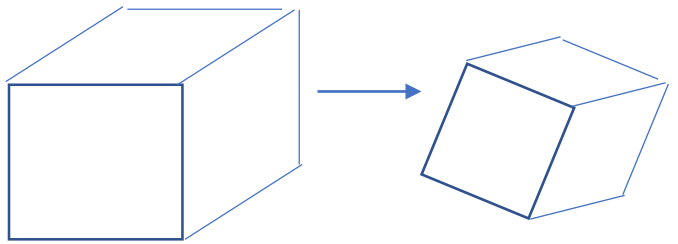
$R_\perp$ is a 3x3 orthogonal matrix: ${R_\perp}^{-1} = {R_\perp}^{T}$

7 dofs: rigid displacement + *scale*

Invariants: ratio of lenghts, angles → shape (not size)

the absolute conic Ω_∞

(*and the absolute dual quadric* ${\boldsymbol{Q}^*}_\infty$)



Affinities (or affine mappings)

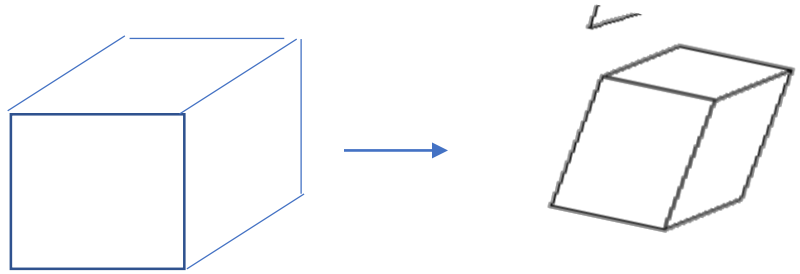
$$H_A = \begin{bmatrix} & A & & \boldsymbol{t} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

A is any 3x3 invertible matrix

12 dofs: A + $\boldsymbol{t}$

Invariants: parallelism, ratio of parallel lenghts, ratio of areas

the plane at the infinity $\boldsymbol{\pi}_\infty$



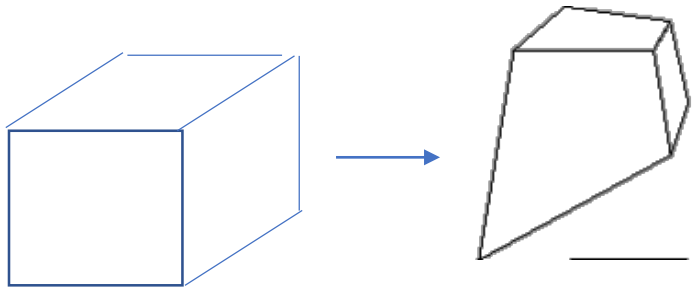
Projectivities (or projective mappings)

$$H = \begin{bmatrix} A & \boldsymbol{t} \\ \boldsymbol{v}^T & 1 \end{bmatrix}$$

A is any 3x3 invertible matrix

15 dofs: $A + \boldsymbol{v} + \boldsymbol{t}$

Invariants: colinearity, incidence, order of contact (crossing, tangency, inflections), the 1D cross ratio, the 2D cross ratio, the 3D cross ratio



3D Shape reconstruction:

will be studied in Multi-view Geometry (next sessions)